

Taming Aleatoric Impulse in Off-Policy Reinforcement Learning

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Abstract

Off-policy reinforcement learning is vulnerable to overestimation bias, which is rooted in the total value uncertainty. However, existing methods typically misaddress this by targeting the epistemic component, neglecting the aleatoric component. We identify for the first time that this oversight fails to contain a massive bias surge, termed **Aleatoric Impulse**. Although transient, this impulse fundamentally derails the learning trajectory, permanently locking the agent into sub-optimal policies. To counteract this, we propose **Aleatoric Impulse Damping (AID)**, the first mechanism that models total value uncertainty by disentangling the return variance into epistemic and aleatoric components, followed by their adaptive weighted recombination. Leveraging this derived uncertainty, the critic constructs a pessimistic lower confidence bound to surgically suppress the impulse. Complementing this, the actor utilizes a symmetrical upper confidence bound to drive optimistic exploration, ensuring that the necessary pessimism does not compromise exploration efficiency. We integrate this mechanism into the Distributional Soft Actor-Critic algorithm to establish **DSAC-AID**. Extensive experiments on the high-dimensional Gym-MuJoCo and DeepMind Control Suite benchmarks demonstrate that it achieves state-of-the-art results in final performance.

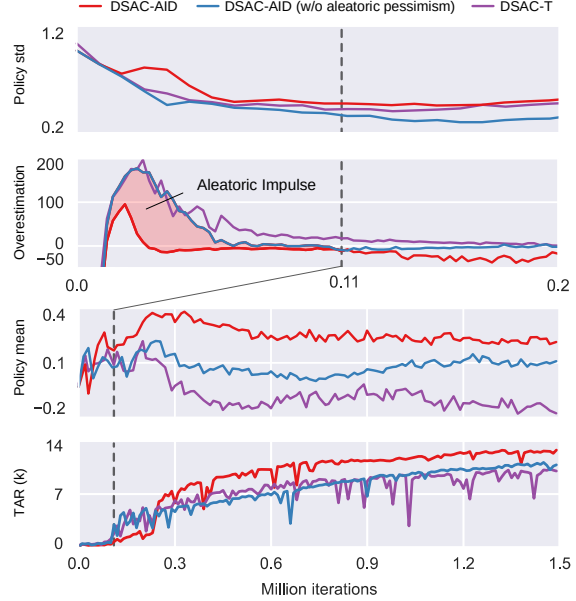


Figure 1. Anatomy and consequence of aleatoric impulse on Humanoid-v3 over one seed. (Top two) The causal link between stochasticity and bias during the early phase (0 – 0.2M). High standard deviation of the policy output amplifies aleatoric uncertainty, triggering a massive aleatoric impulse in value estimation (red shaded) for the unrectified baselines: DSAC-AID w/o aleatoric pessimism ($\beta = 0$) and DSAC-T. Crucially, DSAC-AID ($\beta_{\text{init}} = 0.2$) successfully tames this bias while preserving higher exploration, preventing premature convergence. (Bottom two) Long-term policy evolution and performance (0 – 1.5M). The connecting lines map the early impulse to the onset of irreversible policy divergence. The impulse traps baselines in suboptimal behaviors, whereas DSAC-AID maintains a healthy trajectory, achieving a higher total average return.

1. Introduction

Deep reinforcement learning (DRL) has achieved remarkable success across complex domains, from mastering games (Silver et al., 2017; Vinyals et al., 2019) to robotic control (Rudin et al., 2022; Duan et al., 2024). However, the

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reliability of off-policy DRL is persistently undermined by *value overestimation* (Thrun & Schwartz, 2014), which encourages agents to prioritize overestimated actions over optimal ones, frequently trapping policies in local optima. This bias stems from the uncertainty of value estimation, which can be decomposed into two components: *epistemic uncertainty*, which arises from model ambiguity, and *aleatoric uncertainty*, which stems from inherent environmental and policy stochasticity (Kendall & Gal, 2017; Hüllermeier & Waegeman, 2021).

Standard off-policy algorithms, such as TD3 (Fujimoto et al., 2018) and SAC (Haarnoja et al., 2018), primarily mitigate

overestimation via the *Clipped Double Q-learning* (CDQ) mechanism. This technique implicitly penalizes *epistemic uncertainty* by taking the minimum between two value estimates (see analysis in Appendix B.1). However, this strategy remains blind to the aleatoric uncertainty. Consequently, we identify a critical, yet frequently overlooked phenomenon: **Aleatoric Impulse**. As visualized in Figure 1, our ablation study reveals that it is explicitly triggered by the high policy stochasticity during the early training phase. Although transient, this statistical impulse of overestimation induces a behavioral impulse, compelling the agent to blindly reinforce actions that accidentally produce high returns due to aleatoric noise. If unaddressed, this impulse can fundamentally mislead policy optimization, trapping agents in local optima before effective learning can even begin.

Advancing to distributional reinforcement learning, existing methods remain structurally vulnerable against this impulse. While approaches such as DSAC-T (Duan et al., 2025) and IV-RL (Mai et al., 2022) implicitly encapsulate the critical *aleatoric uncertainty* within the learned *total return uncertainty* (manifested as distribution variance), they fail to isolate it for target rectification. Instead, this information is confined solely to gradient modulation. We argue that such indirect mitigation, akin to network regularization in BRO (Nauman et al., 2024), merely attenuates noise propagation but fails to rectify the inflated targets at their source. In contrast, DoubleGum (Hui et al., 2023) enforces explicit penalization but misdirects it towards the *total return uncertainty*. Since aleatoric impulse manifests as a surge specifically in the *total value uncertainty*, this misalignment renders the suppression ineffective.

To fully resolve this, we propose **Aleatoric Impulse Damping (AID)** mechanism integrated within the Distributional Soft Actor-Critic architecture. The resulting **DSAC-AID** algorithm systematically tames aleatoric impulse through a cohesive uncertainty management system, distinctively coordinated across the critic and actor: (1) The critic leverages a distributional ensemble to explicitly disentangle epistemic and aleatoric uncertainties from the modeled return distributions. On one path, they are aggregated via an adaptive weighted aggregation into the *total value uncertainty* to construct a pessimistically rectified target, surgically suppressing aleatoric impulse during the early training stage. On the parallel path, they are recombined into the *total return uncertainty* to modulate gradient updates via the distributional loss. (2) The actor leverages the obtained *total value uncertainty* to construct an optimistic exploration objective. By maintaining an auxiliary policy driven by this optimism, it ensures that the critic can inject sufficient pessimism to damp the impulse, while the actor remains emboldened to explore potentially optimal regions.

In this paper, we introduce two key contributions: (1) We

identify **Aleatoric Impulse** for the first time, a critical yet previously overlooked phenomenon characterized by a significant surge in value bias during the early training phase. We reveal that this impulse induces irreversible policy divergence, fundamentally limiting the asymptotic performance of existing off-policy algorithms. (2) We propose a novel **AID** mechanism to precisely tame this impulse. Diverging from prior methods that rely on epistemic uncertainty, AID establishes the first framework that explicitly models the *total value uncertainty* by disentangling *total return uncertainty* into epistemic and aleatoric components, followed by their adaptive weighted recombination. Leveraging this regulated uncertainty, we construct adaptive confidence bounds where the pessimistic lower bound constrains the critic to suppress the impulse while the optimistic upper bound guides the actor to neutralize the conservative damping, ultimately reconciling the conflict between robust stability and efficient exploration.

2. Preliminaries

2.1. Problem Formulation

We consider a standard reinforcement learning setting formulated as a Markov Decision Process (MDP) $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$, where $s \in \mathcal{S}$ and $a \in \mathcal{A}$ represent the state and action, respectively (Li, 2023). The dynamics are governed by the transition probability $p(s'|s, a)$ and the reward function $r(s, a)$. The agent aims to learn a policy $\pi(a|s)$ that maximizes the maximum entropy objective: $J(\pi) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t (r_t + \alpha \mathcal{H}(\pi(\cdot|s_t)))]$, where α is the temperature parameter controlling the stochasticity of the optimal policy.

2.2. Distributional Soft Actor-Critic

The Distributional Soft Actor-Critic (DSAC) algorithm (Duan et al., 2021) is an off-policy method that learns the full distribution of the state-action return, parameterized as a Gaussian $Z_{\theta}(\cdot|s, a) = \mathcal{N}(Q_{\theta}(s, a), \sigma_{\theta}^2(s, a))$, where θ denotes the parameters of the critic network. By minimizing the Kullback-Leibler (KL) divergence between the distributional Bellman target $\mathcal{T}Z(s, a) = r + \gamma Z(s', a')$ and the parametric distribution Z_{θ} , the gradient update for the mean estimator Q_{θ} is inversely modulated by the estimated variance σ_{θ}^2 . This variance-aware optimization ensures that Q_{θ} strictly converges to the true expected return.

To further suppress overestimation and enhance robustness, DSAC-T (Duan et al., 2025) incorporates a twin value distribution learning mechanism. The algorithm maintains two independent critics parameterized by θ_1 and θ_2 , and an actor network parameterized by ϕ , along with their corresponding target networks parameterized by $\bar{\theta}_1$, $\bar{\theta}_2$, and $\bar{\phi}$. These target networks are updated via Polyak averaging to stabilize

training. During the update, DSAC-T utilizes the critic with the smaller expected value to construct a conservative target. Let $\bar{i} = \arg \min_{i=1,2} Q_{\bar{\theta}_i}(s', a')$ be the index of the target critic with the lower mean. The targets for the mean (y_q) and variance (y_z) updates are decoupled as follows:

$$y_q = r + \gamma(Q_{\bar{\theta}_i}(s', a') - \alpha \log \pi_{\bar{\phi}}(a'|s')), \quad (1)$$

$$y_z = r + \gamma(z'_{\bar{\theta}_i} - \alpha \log \pi_{\bar{\phi}}(a'|s')), \quad (2)$$

where $a' \sim \pi_{\bar{\phi}}(\cdot|s')$ and $z'_{\bar{\theta}_i} \sim Z_{\bar{\theta}_i}(\cdot|s', a')$. The critic parameters θ_i (for $i = 1, 2$) are updated by minimizing the divergence between the current and target distributions. The gradients are explicitly derived as follows:

$$\begin{aligned} \nabla_{\theta_i} J_Z \approx \mathbb{E}_{\mathcal{B}} \left[-\frac{\mathbb{E}_{\mathcal{B}}[\sigma_{\theta_i}^2]}{\sigma_{\theta_i}^2} (y_q - Q_{\theta_i}) \nabla Q_{\theta_i} \right. \\ \left. - \frac{\mathbb{E}_{\mathcal{B}}[\sigma_{\theta_i}^2]}{\sigma_{\theta_i}^2} \frac{(y_z - Q_{\theta_i})^2 - \sigma_{\theta_i}^2}{\sigma_{\theta_i}} \nabla \sigma_{\theta_i} \right], \end{aligned} \quad (3)$$

where $\mathcal{B}$ denotes the replay buffer, and the introduced scaling term $\mathbb{E}_{\mathcal{B}}[\sigma_{\theta_i}^2]$ serves to enhance training stability.

Subsequently, the policy π_{ϕ} is updated to maximize the expected return estimated by the twin critics:

$$J_{\pi}(\phi) = \mathbb{E}_{s \sim \mathcal{B}, a \sim \pi_{\phi}} \left[\min_{i=1,2} Q_{\theta_i}(s, a) - \alpha \log \pi_{\phi}(a|s) \right]. \quad (4)$$

Finally, to balance exploration and exploitation automatically, the temperature parameter α is adjusted to satisfy a target entropy constraint $\bar{\mathcal{H}}$ by minimizing the following objective:

$$J(\alpha) = \mathbb{E}_{a \sim \pi_{\phi}} [-\alpha \log \pi_{\phi}(a|s) - \alpha \bar{\mathcal{H}}]. \quad (5)$$

3. Aleatoric Impulse

Aleatoric impulse is a transient overestimation surge rooted in the total value uncertainty. By delineating the Q-value hierarchy and decomposing the uncertainty into epistemic and aleatoric components, we reveal how high policy stochasticity and sparse data during early training amplify aleatoric uncertainty to trigger this impulse.

3.1. Hierarchy of Value Estimations

The true Q-value is inherently deterministic, whereas its estimator involves epistemic and aleatoric uncertainty (Chapentier et al., 2022). To clarify the sources of this uncertainty, we identify three conceptual levels:

1. **True Q-value:** This is the theoretical ground truth, defined as the expected return under the current policy π . It is a deterministic scalar representing the perfectly accurate objective:

$$\tilde{Q}(s, a) = \mathbb{E}[Z(s, a)]. \quad (6)$$

2. **Posterior Q-value (finite networks, infinite samples):** Approximating the true value $\tilde{Q}$ with parameters θ on an infinite replay buffer $\mathcal{B}$ does not yield a unique solution. Instead, random initializations and non-convex optimization landscapes drive networks toward diverse local optima. This induces an implicit posterior $\theta \sim p(\theta|\mathcal{B})$, representing the **epistemic uncertainty** inherent to the optimization process. Ideally, robust valuation requires integrating over this posterior:

$$\tilde{Q}(s, a) = \mathbb{E}_{\theta}[\mathbb{E}[Z(s, a; \theta)]]. \quad (7)$$

To approximate this intractable integral, practical algorithms (Lee et al., 2021; Li et al., 2023; Cetin & Celiktutan, 2023) use an ensemble $\{\theta_i\}_{i=1}^m$ to estimate the value as the mean of individual expectations:

$$Q(s, a) = \frac{1}{m} \sum_{i=1}^m \mathbb{E}[Z(s, a; \theta_i)], \quad (8)$$

where the expectation term $\mathbb{E}[Z(s, a; \theta_i)]$ corresponds to the parametric value estimate $Q_{\theta_i}(s, a)$.

3. **Empirical Q-value (finite networks, finite samples):** In practice, trained on finite samples from the replay buffer, the neural network’s estimation can be modeled as an arithmetic mean of observed returns. For a specific parameter realization θ_i , this is formulated as:

$$Q(s, a) \approx \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n z_{ij}(s, a; \theta_i), \quad (9)$$

where n denotes the effective sample count contributing to the estimate, and $z_{ij} \sim Z_{\theta_i}(s, a)$ represents the stochastic value sample that encapsulates the inherent randomness arising from environmental dynamics and policy stochasticity. Thus, while the irreducible noise in individual samples z_{ij} is theoretically suppressed by this averaging process, the residual noise that persists due to finite effective sample support constitutes the **aleatoric uncertainty** of the estimator.

3.2. Decomposition of Value Uncertainty

Following the hierarchy analysis, we decompose the total value uncertainty into epistemic and aleatoric components using the *Law of Total Variance*. The detailed mathematical proof of this decomposition is provided in Appendix A.2.

$$\text{Var } Q = \underbrace{\text{Var}_{\theta}[\mathbb{E}Q_{\theta}]}_{\text{Epistemic}} + \underbrace{\mathbb{E}_{\theta}[\text{Var } Q_{\theta}]}_{\text{Aleatoric}}. \quad (10)$$

While direct quantification of the aleatoric component is intractable (see justification in Appendix B.3), it is intrinsically linked to the variance of the underlying return distribution Z . Through the derivation detailed in Appendix A.3,

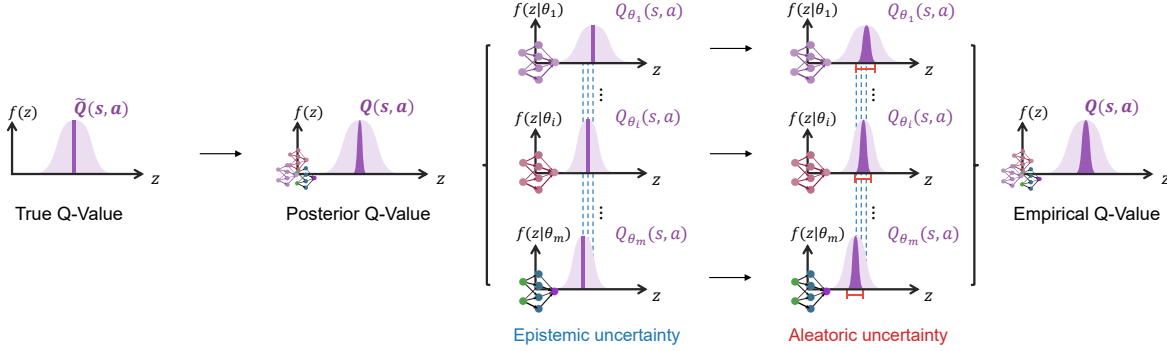


Figure 2. The hierarchy of value estimations and uncertainty decoupling. The schematic illustrates the transition from the theoretical true value to the empirical ensemble. The light purple shaded areas depict return distributions, while dark purple elements represent Q-values. Crucially, this ensemble architecture explicitly decomposes the total uncertainty into two distinct components: epistemic uncertainty, which is quantified by the divergence among the predicted means of the individual ensemble members (indicated by the blue dashed lines); and aleatoric uncertainty, which is captured by the intrinsic variance of the individual distributions (indicated by the red horizontal bars). Finally, the resulting empirical Q-value aggregates these estimations, representing a final distribution where the total uncertainty encompasses both the epistemic and aleatoric components.

we establish that the aleatoric value uncertainty is a scaled manifestation of the aleatoric return uncertainty:

$$\text{Var } Q \approx \underbrace{\text{Var}_{\theta} Q_{\theta}}_{\text{Epistemic}} + \underbrace{(1/n) \mathbb{E}_{\theta}[\text{Var } Z_{\theta}]}_{\substack{\text{Aleatoric uncertainty of } Z \\ \text{Aleatoric uncertainty of } Q}}, \quad (11)$$

where $1/n$ serves as a scaling factor, bridging the aleatoric uncertainty of the return estimation to that of the value estimation. Intuitively, this relationship suggests that the inherent aleatoric noise is not entirely filtered out but is suppressed by the effective sample count n available to the estimator.

3.3. Emergence of Aleatoric Impulse

Prior research (Duan et al., 2021) establishes that the overestimation bias induced by the maximization operator scales with the standard deviation of the value estimation (see justification in Appendix A.1). Substituting (11) into this relationship implies that the total bias is a composite of both epistemic and aleatoric uncertainties. This decomposition provides the theoretical basis for aleatoric impulse. Specifically, it is the aleatoric component that acts as the dominant driver of the bias surge during the initial training phase. To characterize this impulse, we proceed by addressing two critical questions.

Why does aleatoric uncertainty explode during early training? We attribute this phenomenon to a critical resonance between two factors. First, the initial scarcity of data constrains the effective sample count, forcing the weights to concentrate on a few available neighbors. As analyzed in Appendix A.4, this sustains the coefficient $1/n$ at a critically high level, providing an unimpeded channel for noise propagation. Second, this period of maximal sensitivity co-

incides with the high stochasticity of early-stage policies (as visualized in Figure 1). Since these policies are initialized to explore, they inherently generate returns with high aleatoric uncertainty. Consequently, the maximum inherent stochasticity is fully transmitted through the unattenuated weighting factor, triggering the intense surge of overestimation bias.

Why does this transient impulse jeopardize long-term learning? Since $1/n$ naturally decays as data accumulates, one might assume that this initial impulse is negligible. However, we contend that it is destructive due to bootstrapping. If unaddressed, the intense bias injected into early value estimates misleads the agent to overcommit to suboptimal actions with falsely inflated value. This derails the learning trajectory, permanently trapping the agent in local optima even long after original aleatoric impulse has subsided. Thus, explicit damping is essential to aid the agent through this critical initial phase.

4. Aleatoric Impulse Damping

Building on the theoretical analysis in Sec. 3, we propose **DSAC-AID** (Distributional Soft Actor-Critic with Aleatoric Impulse Damping). This framework systematically tames aleatoric impulse by integrating pessimistic evaluation in the critic and optimistic exploration in the actor.

4.1. Pessimistic Evaluation

The critic maintains an ensemble of m parametric return distributions defined by $\{\theta_i\}_{i=1}^m$. Each takes the form of a Gaussian $Z_{\theta_i} \sim \mathcal{N}(Q_{\theta_i}, \sigma_{\theta_i}^2)$, where Q_{θ_i} estimates the expected action value and σ_{θ_i} captures the inherent aleatoric return uncertainty. Leveraging this ensemble structure, we derive the aggregated mean value as: $Q(s, a) = \frac{1}{m} \sum_{i=1}^m Q_{\theta_i}(s, a)$. Simultaneously, we isolate epistemic uncertainty σ_{epi} and

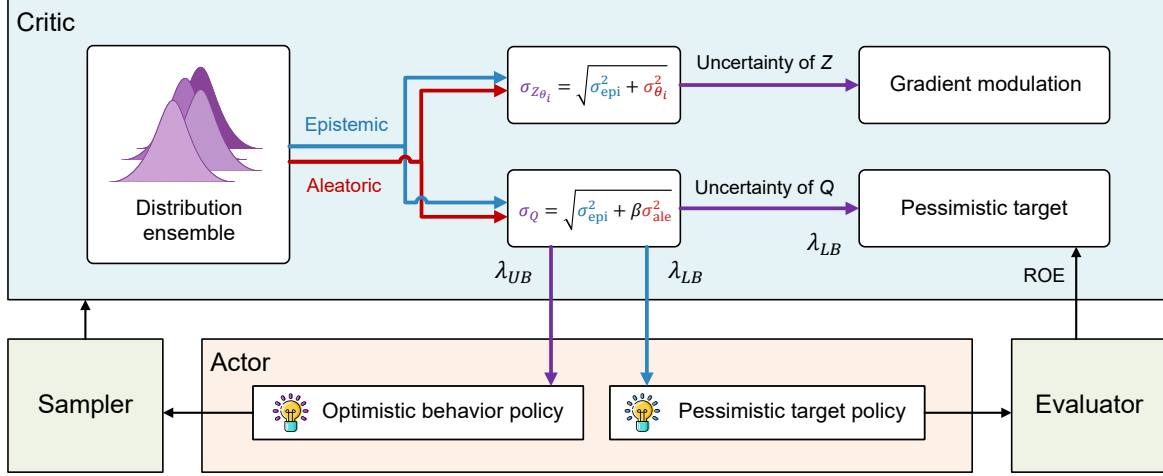


Figure 3. **The framework of DSAC-AID.** The architecture mitigates aleatoric impulse through two mechanisms: (1) **Pessimistic Evaluation:** The critic decouples uncertainty sources to enforce targeted suppression and gradient modulation, with the coefficient β adaptively regulated via relative overestimation error. (2) **Optimistic Exploration:** The behavior policy utilizes the total value uncertainty as an exploration bonus to symmetrically counterbalance the conservative critic.

aleatoric uncertainty σ_{ale}^2 , which are formally quantified as:

$$\sigma_{\text{epi}}^2(s, a) = \frac{1}{m-1} \sum_{i=1}^m [Q_{\bar{\theta}_i}(s, a) - Q(s, a)]^2, \quad (12)$$

$$\sigma_{\text{ale}}^2(s, a) = \frac{1}{m} \sum_{i=1}^m \sigma_{\bar{\theta}_i}^2(s, a). \quad (13)$$

Based on these decoupled metrics, we construct a pessimistic target and implement a gradient modulation scheme to stabilize learning.

Pessimistic target. Aligning with our theory, we construct a lower confidence bound based on total value uncertainty to rectify the overestimation bias. This formulation explicitly integrates the aleatoric component with the epistemic one to effectively damp aleatoric impulse. The enhanced critic target is defined as:

$$y_q = r + \gamma [Q(s', a') - \lambda_{LB} \sigma_Q(s', a') - \alpha \log \pi_{\bar{\phi}}(a'|s')], \quad (14)$$

where the confidence coefficient λ_{LB} provides controllable pessimism. The total value uncertainty σ_Q is synthesized as

$$\sigma_Q(s', a') = \sqrt{\sigma_{\text{epi}}^2(s', a') + \beta \sigma_{\text{ale}}^2(s', a')}. \quad (15)$$

The coefficient β governs the suppression of aleatoric impulse. In practice, we employ a decoupled formulation to ensure β acts as an independent parameter (Appendix C.1). To mirror the theoretical decay of $1/n$, β anneals from a high initial value to zero. In particular, it reduces to the standard CDQ formulation in the limit under the specific setting of $\lambda_{LB} = 1/\sqrt{2}$ and $m = 2$. Accordingly, we automate this annealing via a closed-loop mechanism based on the

relative overestimation error (ROE). Specifically, we compute the raw estimation bias using on-policy Monte Carlo returns (Appendix C.2) and normalize the average bias of the first K state-action pairs by the estimated uncertainty to constitute the relative metric:

$$\beta \leftarrow \beta + l_\beta \frac{\min(0, \frac{1}{K} \sum_{t=1}^K e_t(s_t, a_t))}{\mathbb{E}_{(s,a) \sim \mathcal{B}}[\sigma_Q(s, a)]}, \quad (16)$$

where e_t denotes the raw bias of the k -th state-action pair, and l_β denotes the annealing rate. The $\min(0, \cdot)$ operator ensures that β is reduced only when underestimation is detected. This prevents the mechanism from interfering with late-stage overestimation, which is typically driven by epistemic uncertainty and handled by λ_{LB} . Computationally, this process leverages trajectories generated for the total average return calculation, incurring negligible overhead.

Gradient modulation. The optimization of distributional value estimation inherently scales gradients by the inverse of the learned variance $\sigma_{\bar{\theta}_i}^2$. In standard approaches like DSAC-T, the reliance on a shared target compels the learned variance to capture the total return uncertainty, which inherently entangles both aleatoric and epistemic components (see derivation in Appendix B.2). To ensure the model learns a clean and accurate aleatoric distribution, $\sigma_{\bar{\theta}_i}$ must track an independent sub-target y_{z_i} :

$$y_{z_i} = r + \gamma (z'_{\bar{\theta}_i} - \alpha \log \pi_{\bar{\phi}}(a'|s'))|_{z'_{\bar{\theta}_i} \sim Z_{\bar{\theta}_i}(\cdot|s', a')}. \quad (17)$$

Since this target limits $\sigma_{\bar{\theta}_i}$ to capture only aleatoric uncertainty, the standard inverse-variance scaling fails to dampen value updates in the presence of epistemic uncertainty. To enforce conservatism against out-of-distribution samples, we explicitly incorporate the epistemic uncertainty into the

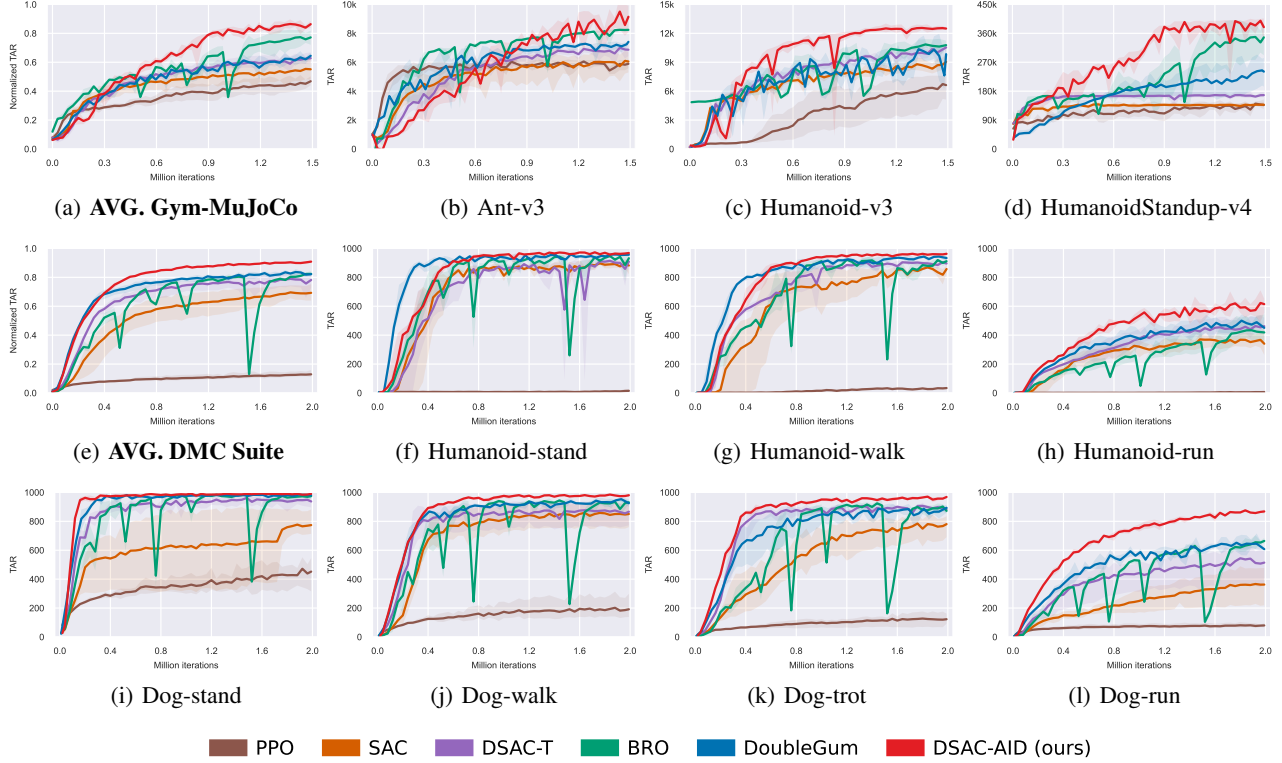


Figure 4. Training curves of model-free algorithms on benchmarks. The solid lines represent the mean, while the shaded regions indicate the confidence interval over 3 seeds. The average performance curves appear at the left of the 1st and 2nd rows, highlighted in **bold**. Note that for both aggregate plots, the scores of individual benchmarks are normalized by task-specific thresholds before averaging. Additionally, BRO employs 1 environment interaction per step and an update-to-data (UTD) ratio of 10, whereas other algorithms utilize 20 environment interactions per step and a UTD ratio of 1.

gradient scaler for the mean Q_{θ_i} . The resulting modulated gradients are formulated as:

$$\nabla_{\theta_i} J_Z \approx \mathbb{E}_B \left[-\frac{\mathbb{E}_B[\sigma_{\theta_i}^2 + \sigma_{\text{epi}}^2]}{\sigma_{\theta_i}^2 + \sigma_{\text{epi}}^2} (y_q - Q_{\theta_i}) \nabla Q_{\theta_i} - \frac{\mathbb{E}_B[\sigma_{\theta_i}^2]}{\sigma_{\theta_i}^2} \frac{(y_{z_i} - Q_{\theta_i})^2 - \sigma_{\theta_i}^2}{\sigma_{\theta_i}} \nabla \sigma_{\theta_i} \right], \quad (18)$$

where $\sigma_{\theta_i}^2 + \sigma_{\text{epi}}^2$ estimates the uncertainty of Z_{θ_i} . In contrast, the update for σ_{θ_i} remains scaled by its own output. It ensures that the learning of aleatoric uncertainty is not hindered by epistemic uncertainty, allowing the model to swiftly capture the true noise distribution even in early training stages.

4.2. Optimistic Exploration

To empower the critic with the sufficient pessimism necessary to completely eliminate aleatoric impulse, the accompanying risk of excessive underestimation must be proactively neutralized. Otherwise, it will inevitably force the policy to retreat into low-uncertainty action zones, abandoning high-potential regions. To counterbalance this, we decouple the actor into a target policy π_T for stable value backup and a

behavior policy π_B for active exploration, parameterized by ϕ_T and ϕ_B .

For π_T , we aim to achieve risk neutrality with respect to inherent stochasticity while ensuring robustness against model ambiguity. Thus, π_T maximizes the Q-value penalized solely by epistemic uncertainty:

$$J_{\pi_T}(\phi_T) = \mathbb{E}[Q - \lambda_{\text{LB}} \sigma_{\text{epi}} - \alpha \log \pi_{\phi_T}]. \quad (19)$$

In contrast, π_B is designed to strictly counterbalance the critic's suppression. It incorporates an upper confidence bound based on the total value uncertainty:

$$J_{\pi_B}(\phi_B) = \mathbb{E}[Q + \lambda_{\text{UB}} \sigma_Q - \alpha \log \pi_{\phi_B}], \quad (20)$$

where the confidence coefficient λ_{UB} controls optimism. Similar to the critic, we apply structural decoupling to σ_Q here to ensure exact aleatoric neutralization (see Appendix C.1). Crucially, since σ_Q inherently contains the aleatoric coefficient β , this optimism symmetrically counterbalances the critic's pessimism, ensuring precise compensation for the aleatoric suppression.

Table 1. Total Average Return (TAR) of model-free algorithms on DMC Suite and Gym-MuJoCo tasks. Mean $\pm$ Std over 3 seeds (12345, 22345, 32345). Gym-MuJoCo values are in k (10^3). Darker Blue = best, Lighter Blue = second-best. The average scores of individual benchmarks ($\dagger$) are normalized by task-specific thresholds before averaging: Ant (10k), Humanoid (15k), and HumanoidStandup (450k) for Gym-MuJoCo; all DMC Suite tasks are normalized by 1k.

Task	PPO	SAC	DSAC-T	BRO	DoubleGum	DSAC-AID (ours)
Ant-v3 (k)	6.2 $\pm$ 0.2	6.4 $\pm$ 0.8	7.1 $\pm$ 0.3	8.3 $\pm$ 0.1	7.5 $\pm$ 0.0	10.0 $\pm$ 0.4
Humanoid-v3 (k)	6.9 $\pm$ 1.6	9.4 $\pm$ 0.6	10.8 $\pm$ 0.2	11.0 $\pm$ 0.3	10.8 $\pm$ 0.9	12.7 $\pm$ 0.3
HumanoidStandup-v4 (k)	147.0 $\pm$ 5.7	140.5 $\pm$ 5.1	168.3 $\pm$ 7.3	368.4 $\pm$ 46.0	252.5 $\pm$ 37.0	405.2 $\pm$ 1.9
AVG. Gym-MuJoCo†	0.469 $\pm$ 0.135	0.527 $\pm$ 0.161	0.602 $\pm$ 0.162	0.793 $\pm$ 0.074	0.677 $\pm$ 0.101	0.918 $\pm$ 0.070
Dog-stand	479.3 $\pm$ 85.6	790.4 $\pm$ 77.3	966.5 $\pm$ 25.4	993.6 $\pm$ 4.0	992.7 $\pm$ 1.8	994.6 $\pm$ 0.3
Dog-walk	210.1 $\pm$ 74.8	869.4 $\pm$ 67.2	889.1 $\pm$ 81.3	964.2 $\pm$ 1.7	965.4 $\pm$ 7.9	991.3 $\pm$ 1.9
Dog-trot	133.3 $\pm$ 44.6	794.2 $\pm$ 72.0	921.6 $\pm$ 9.7	944.7 $\pm$ 2.2	907.2 $\pm$ 25.1	982.4 $\pm$ 3.8
Dog-run	85.6 $\pm$ 13.8	372.4 $\pm$ 108.0	558.0 $\pm$ 124.9	687.4 $\pm$ 15.9	699.6 $\pm$ 37.1	884.2 $\pm$ 4.4
Humanoid-stand	15.6 $\pm$ 9.3	907.5 $\pm$ 15.9	933.3 $\pm$ 2.0	961.8 $\pm$ 4.7	966.7 $\pm$ 3.2	976.1 $\pm$ 1.1
Humanoid-walk	38.5 $\pm$ 3.2	881.5 $\pm$ 2.5	920.1 $\pm$ 0.7	940.3 $\pm$ 6.7	951.1 $\pm$ 5.7	967.2 $\pm$ 0.8
Humanoid-run	6.8 $\pm$ 3.7	394.9 $\pm$ 46.9	473.0 $\pm$ 49.0	472.4 $\pm$ 66.3	523.6 $\pm$ 42.1	685.9 $\pm$ 28.3
AVG. DMC Suite†	0.138 $\pm$ 0.161	0.716 $\pm$ 0.224	0.809 $\pm$ 0.197	0.853 $\pm$ 0.181	0.858 $\pm$ 0.166	0.926 $\pm$ 0.105

5. Experiments

5.1. Experimental Setup

Baselines. We evaluate our method against nine representative algorithms categorized into two paradigms: (1) model-free baselines, including classic PPO (Schulman et al., 2017) and SAC (Haarnoja et al., 2018), and state-of-the-art distributional methods DSAC-T (Duan et al., 2025), DoubleGum (Hui et al., 2023), and BRO (Nauman et al., 2024); and (2) model-based baselines, comprising state-of-the-art off-policy approaches DreamerV3 (Hafner et al., 2025), TD-MPC2 (Hansen et al., 2024), BMPC (Wang et al., 2025), and BOOM (Zhan et al., 2025). The detailed implementation for all algorithms is documented in Appendix D.2.

Benchmarks. DSAC-AID is evaluated on a challenging benchmark of 10 high-dimensional locomotion tasks drawn from MuJoCo (Todorov et al., 2012) continuous control suite in OpenAI Gym (Brockman et al., 2016) and the DeepMind Control Suite (DMC) (Tassa et al., 2018). The selected 3 Gym-MuJoCo tasks and 7 DMC tasks feature three of the most complex agents, including humanoid, ant, and dog. These tasks demand sophisticated balance and coordination in high-dimensional observation and continuous action spaces. Detailed descriptions are listed in Appendix D.1.

5.2. Experimental Results

We detail the performance against model-free baselines in Figure 4 and Table 1, while deferring the extensive model-based comparisons to Appendix E.1. Our method, DSAC-AID, consistently delivers the best total average return (TAR) across all high-dimensional locomotion tasks, demonstrating remarkable stability even in environments with complex contact dynamics and large state-action spaces.

Results on Gym-MuJoCo. DSAC-AID achieves a normalized TAR of 0.918, a **+15.8%** improvement over the strongest baseline BRO (0.793). Notably, it achieves significant margins in Ant-v3 (10.0k, **+20.5%**) and Humanoid-v3 (12.7k, **+15.5%**). Furthermore, in HumanoidStandup-v4, it reaches a high return of 405.2k, demonstrating robust performance in handling complex dynamics.

Results on DMC Suite. DSAC-AID achieves an average TAR of 0.926, outperforming the most competitive baselines, including model-free DoubleGum (**+7.9%**) and model-based BOOM (**+5.5%**). The advantage is particularly pronounced in complex tasks: in Humanoid-run, it leads BOOM by **+17.7%** and DoubleGum by **+31.0%**, while also surpassing the leading model-based method in Dog-run (**+7.7%**). Crucially, DSAC-AID attains these results without the inference-time computational cost inherent to model-based approaches.

5.3. Overestimation Analysis

Figure 1 and Figure 5 provide a direct visualization of the value overestimation bias, offering empirical evidence for the damping effect of our mechanism (see Appendix E.3 for extended plots). During the early stage, objective-regularized algorithms (SAC, DSAC-T) suffer from a severe aleatoric impulse. While network-regularized baselines (BRO, DoubleGum) reduce the average bias, they fail to suppress the overestimation ratio consistently. This suggests that network regularization functions as a low-pass filter, dampening high-frequency epistemic noise while leaving low-frequency aleatoric surges unconstrained. In contrast, DSAC-AID maintains the lowest ratio throughout the entire training course. The observed residual peaks are attributable to structural discrepancies inherent to the bias metric itself, as detailed in Appendix C.2.

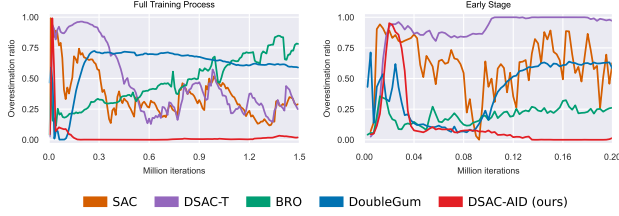


Figure 5. Analysis of overestimation ratio on Humanoid-v3. We compare DSAC-AID against model-free baselines over the full training process and the early stage.

5.4. Ablation Study

We conducted ablation studies to validate the efficacy of the two key components and the sensitivity of two key hyperparameters.

Component Efficacy. We quantify the contributions of distinct algorithmic modules on two representative tasks, as detailed in Figure 6. First, eliminating the aleatoric pessimism (by setting $\beta_{\text{init}} = 0$) degrades performance by 15.2% on Humanoid-v3 and 9.7% on Humanoid-run, highlighting its foundational role in stabilizing value learning. Furthermore, we find that the annealing of this pessimism is equally critical. Eliminating the pessimism annealing (by setting $l_\beta = 0$) leads to severe training instability and a drastic performance collapse on Humanoid-v3, alongside a significant performance drop of 10.8% on Humanoid-run.. This empirical finding strongly corroborates our theoretical analysis: since the aleatoric uncertainty of the value estimate naturally vanishes as data accumulates, a fixed penalty introduces excessive conservatism in the late stage, confirming the necessity of the pessimism annealing. Second, discarding the optimistic exploration module (by setting $\lambda_{\text{UB}} = 0$) results in deficits of 11.4% and 14.8% respectively, hindering the discovery of high-reward regions. These results confirm that combining these components is essential for mastering high-dimensional tasks.

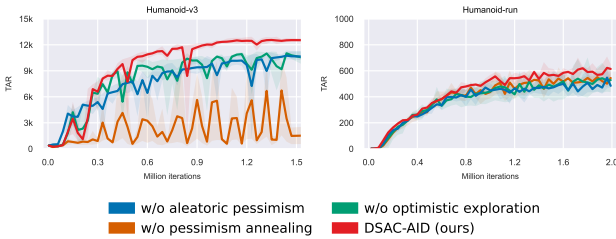


Figure 6. Ablation study on component efficacy. Performance comparison on Gym-MuJoCo Humanoid-v3 and DMC Humanoid-run.

Hyperparameter sensitivity. We evaluate the robustness of DSAC-AID regarding the initial pessimism coefficient β_{init} and its annealing rate l_β . As shown in Figure 7, our method maintains high performance across a broad range

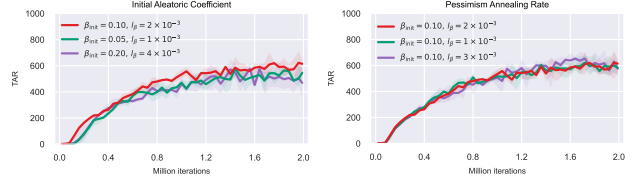


Figure 7. Hyperparameter sensitivity analysis on Humanoid-run. We evaluate the impact of varying the initial pessimism coefficient β_{init} and its annealing rate l_β .

of settings for both hyperparameters, indicating that DSAC-AID exhibits strong stability without requiring meticulous tuning.

6. Related Work

Existing approaches for mitigating value overestimation in off-policy RL can be broadly categorized into two paradigms: objective and network regularization.

Objective regularization. This paradigm stabilizes value estimation by modifying the learning objective (see Appendix F for a detailed comparison). *Implicit approaches* rely on structural heuristics or distributional constraints. Heuristic methods like TD3 (Fujimoto et al., 2018) impose pessimism via the minimum of value estimates. Alternatively, distributional methods like DSAC-T (Duan et al., 2025) implicitly regularize gradients utilizing the variance of the learned return distribution. *Explicit approaches*, conversely, construct penalties based on quantified uncertainty components. Existing methods typically target a single source: RAC (Li et al., 2023) and GPL (Cetin & Celiktutan, 2023) restrict pessimism to epistemic uncertainty, whereas UA-DQN (Clements et al., 2019) focuses solely on aleatoric uncertainty. While DoubleGum (Hui et al., 2023) incorporates both, it erroneously targets the *total return uncertainty* rather than the *total value uncertainty*. In contrast, DSAC-AID explicitly disentangles these sources to construct a dynamic, closed-loop penalty targeting the precise total value uncertainty.

Network regularization. Recent methods implicitly mitigate overestimation by stabilizing approximation via specific normalization layers, exemplified by LayerNorm (Ba et al., 2016) in BRO (Nauman et al., 2024) and GroupNorm (Wu & He, 2018) in DoubleGum (Hui et al., 2023). Furthermore, Simba (Lee et al., 2025) employs hyperspherical normalization to constrain norm growth. However, heavy reliance on such static structural constraints imposes a constant regularization cost, limiting model expressivity in low-risk regions. In contrast, DSAC-AID employs GroupNorm solely for fundamental stability, relying instead on an adaptive uncertainty penalty to precisely suppress residual bias.

7. Conclusion

We first uncover aleatoric impulse, a phenomenon where environmental and policy stochasticity triggers significant value overestimation during early training. To resolve this, we propose DSAC-AID, the first algorithm to model total value uncertainty by explicitly disentangling the total return uncertainty into epistemic and aleatoric components, followed by their adaptive weighted recombination. Based on this derived metric, DSAC-AID integrates pessimistic evaluation with optimistic exploration, suppressing the impulse without hindering policy improvement to achieve state-of-the-art results. Future work could explore directly estimating total value uncertainty. This approach would streamline the disentanglement process and reduce hyperparameter reliance, enabling a more concise and automated framework.

Our work clarifies the distinct impacts of uncertainty types on value estimation. Crucially, we demonstrate that explicitly modeling the total value uncertainty is essential to rectify the structural neglect of aleatoric uncertainty in existing paradigms. Ultimately, to transform such stochasticity from a source of instability into a tractable signal for robust value estimation, off-policy reinforcement learning fundamentally needs AID.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (NSFC) under grant number 52221005, and the Tsinghua University (State Key Laboratory of Intelligent Green Vehicle and Mobility)–Xiaomi EV Technology Co., Ltd Joint Research Center for Autonomous Driving Safety and Intelligence. The core of this work was completed in the Intelligence Driving Lab (IDLAB) at Tsinghua University. We extend our sincere gratitude to all the lab members for their constructive feedback during this period.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Theoretical Derivation

A.1. Overestimation Bias via Value Uncertainty

In this section, we analyze how value uncertainty explicitly contributes to overestimation bias. Consider the standard Q-learning target $y = \mathbb{E}[r] + \gamma \mathbb{E}_{s'}[\max_{a'} Q_\theta(s', a')]$. The parameters θ are updated to minimize the squared loss between this target and the current estimate.

Due to epistemic and aleatoric uncertainty, the estimated Q-values inevitably contain noise. We model the estimate as the true value $\tilde{Q}$ perturbed by a zero-mean random error term: $Q(s, a) = \tilde{Q}(s, a) + \epsilon_Q$. Crucially, the target calculation involves maximizing over the noisy estimates at the next state s' . We denote the corresponding noise as ϵ'_Q , which follows a distribution $\mathcal{N}(0, \sigma_Q^2)$ where the variance σ_Q^2 reflects the total value uncertainty. Following the analysis in prior work (Duan et al., 2021), the expected one-step overestimation bias b is proportional to the gap between the expected maximum of the noisy values and the maximum of the true values:

$$b[Q(s, a)] \propto \mathbb{E}_{s'} \left[\mathbb{E}_{\epsilon'_Q} \left[\max_{a'} (\tilde{Q}(s', a') + \epsilon'_Q) \right] - \max_{a'} \tilde{Q}(s', a') \right]. \quad (21)$$

According to Jensen’s inequality, the expectation of the convex maximization operator is greater than or equal to the maximum of the expectation, i.e., $\mathbb{E}[\max(\tilde{Q} + \epsilon)] \geq \max(\tilde{Q})$. Equation (21) mathematically demonstrates that the magnitude of this gap scales with the variance σ_Q^2 of the error term ϵ'_Q . High uncertainty implies a larger variance, which inflates the expectation term $\mathbb{E}[\max(\cdot)]$, resulting in significant value overestimation. This theoretical result confirms that explicitly suppressing value uncertainty is essential for mitigating the overestimation bias at its source.

A.2. Decomposition of Total Value Uncertainty

We strictly derive the decomposition of the total uncertainty of the ensemble estimator Q into its aleatoric and epistemic components. By definition, the total variance of Q can be expanded as the difference between the second moment and the square of the first moment. Applying the law of iterated expectations to the ensemble parameters θ , we have:

$$\text{Var } Q = \mathbb{E}[(Q - \mathbb{E}Q)^2] = \mathbb{E}[Q^2] - (\mathbb{E}Q)^2 = \mathbb{E}_\theta[\mathbb{E}[Q_\theta^2]] - (\mathbb{E}_\theta[\mathbb{E}Q_\theta])^2. \quad (22)$$

For a specific realization of parameters θ , the second moment of the value estimate Q_θ is related to its variance and mean by the identity: $\mathbb{E}[Q_\theta^2] = \text{Var } Q_\theta + (\mathbb{E}[Q_\theta])^2$. Substituting this identity into the total variance equation allows us to regroup the terms. By associating the variance of the expectation and the expectation of the variance, we arrive at the final decomposition:

$$\text{Var } Q = \mathbb{E}_\theta[\text{Var } Q_\theta] + \mathbb{E}_\theta[(\mathbb{E}[Q_\theta])^2] - (\mathbb{E}_\theta[\mathbb{E}Q_\theta])^2 = \underbrace{\mathbb{E}_\theta[\text{Var } Q_\theta]}_{\text{Aleatoric}} + \underbrace{\text{Var}_\theta[\mathbb{E}Q_\theta]}_{\text{Epistemic}}. \quad (23)$$

This confirms that the total uncertainty represents the sum of the expected aleatoric uncertainty and the epistemic uncertainty.

A.3. Decomposition of Total Return Uncertainty

Within the ensemble learning framework, the total return $Z(s, a)$ is defined as the uniform mixture of the individual predicted distributions:

$$Z(s, a) = \frac{1}{m} \sum_{i=1}^m Z_{\theta_i}(s, a). \quad (24)$$

Similar to the value estimator, its total variance can be decomposed into epistemic and aleatoric components via the *Law of Total Variance*:

$$\text{Var } Z = \underbrace{\mathbb{E}_\theta[\text{Var } Z_\theta]}_{\text{Aleatoric}} + \underbrace{\text{Var}_\theta[\mathbb{E}Z_\theta]}_{\text{Epistemic}}. \quad (25)$$

Comparing this with the total value uncertainty derived in Eq. (11), we identify the following relationships:

Epistemic consistency. The epistemic component represents the structural disagreement among ensemble members regarding the expected return. Since the Q-value serves as an unbiased estimator of the true expectation $\mathbb{E}[Z_\theta]$, the variance

across models θ targets the underlying mean rather than the sample noise. Consequently, the epistemic uncertainty of the value estimator is equivalent to that of the underlying return distribution:

$$\sigma_{\text{epi}_Q}^2 = \text{Var}_\theta[\mathbb{E}Q_\theta] \approx \text{Var}_\theta \left[\sum_{j=1}^n \frac{1}{n} \mathbb{E}[Z_\theta(s, a)] \right] = \text{Var}_\theta[\mathbb{E}Z_\theta] = \sigma_{\text{epi}_Z}^2. \quad (26)$$

Practically, since Q_θ serves as an estimator for $\mathbb{E}Z_\theta$, we approximate the epistemic uncertainty directly using $\text{Var}_\theta Q_\theta$. This approximation is theoretically justified and holds particularly when $\text{Var} Q_\theta$ is small.

Aleatoric decay. In contrast, the aleatoric component follows the statistical properties of the sample mean. While the aleatoric uncertainty of return $\sigma_{\text{ale}_Z}^2$ is irreducible, its impact on the value estimator is modulated by the effective sample count. By treating the returns Z_θ as random variables, we can formalize this relationship as follows:

$$\sigma_{\text{ale}_Q}^2 = \mathbb{E}_\theta[\text{Var} Q_\theta] \approx \mathbb{E}_\theta \left[\text{Var} \left[\frac{1}{n} \sum_{j=1}^n z_{ij} \right] \right] \stackrel{(a)}{=} \mathbb{E}_\theta \left[\sum_{j=1}^n \frac{1}{n^2} \text{Var}[z_{ij}] \right] \stackrel{(b)}{\approx} \sum_{j=1}^n \frac{1}{n^2} \cdot \mathbb{E}_\theta[\text{Var} Z_\theta] = \frac{1}{n} \cdot \sigma_{\text{ale}_Z}^2. \quad (27)$$

Equality (a) holds because the samples Z_{θ_i} are independently sampled from the distribution $\mathcal{N}(Q_{\theta_i}, \sigma_{\theta_i}^2)$. Approximation (b) holds as the system approaches a steady state, where the estimated return variances $\sigma_{\theta_i}^2$ (which denote $\text{Var} Z_{\theta_i}$) become approximately identical. This stationarity is grounded in the constant nature of environmental noise, the regulated policy entropy in SAC-based frameworks, and the incremental smoothing effect of target networks. Approximation (b) may lead to a slight overestimation of the aleatoric value uncertainty in the initial non-steady state, where the network’s predicted $\sigma_{\theta_i}^2$ is typically increasing. To address this, our algorithm employs a non-increasing constraint on β specified in Equation (16), resulting in the following phases:

- **Early Phase (Non-steady, β remains constant):** During this phase, positive ROE does not trigger a decrease in β . Instead, β remains constant at β_{init} , which is purposefully set lower than the theoretical $1/n$. This choice effectively offsets the temporary overestimation of $\sigma_{\text{ale}_Q}^2$, ensuring that the aleatoric penalty remains stable
- **Late Phase (Steady, β begins to decrease):** As training transitions toward a steady state, positive ROE typically subsides and becomes negative. This triggers the annealing of β , which allows the effective sample count n to grow and precisely reflects the true $1/n$ statistical decay of the estimator’s uncertainty.

A.4. Interpretation of Effective Sample Count

We analyze the effective sample count, denoted as n , which serves as the denominator in our aleatoric uncertainty estimator. Rather than a raw buffer count, n represents the neural network’s local averaging capacity at a query (s, a) , governed by two factors: (1) *Temporal relevance*: Since off-policy TD learning implicitly prioritizes recent updates, effective samples are constrained to a sliding window of recent iterations; and (2) *State-action similarity*: Generalization enables the network to aggregate contributions from a neighborhood of similar latent pairs, rather than relying solely on identical matches. Based on this, we describe the asymptotic behavior of the scaling factor $1/n$:

Early stage. Early training is characterized by scarce historical data and high policy stochasticity, which scatters transitions sparsely across the state space. Consequently, the effective support is minimal, maintaining the scaling factor $1/n$ at a critically high level.

Late stage. As the policy stabilizes and data accumulates, transitions concentrate around optimal trajectories. The effective support becomes dense, causing the scaling factor $1/n \rightarrow 0$. This indicates that the aleatoric uncertainty of the estimator naturally vanishes as the local data support grows sufficiently large.

B. Limitations of Uncertainty Estimation in Prior Works

B.1. Aleatoric Blindness of the Minimum Operator

Widely adopted in prominent algorithms such as TD3 and SAC, the Clipped Double Q-learning (CDQ) paradigm computes the target value via a minimum operator $y = \min(Q_{\theta_1}, Q_{\theta_2})$ to suppress overestimation. This formulation can be

decomposed using the identity $\min(x, y) = \frac{x+y}{2} - \frac{|x-y|}{2}$. Defining the ensemble mean as $\bar{Q} = \frac{Q_{\theta_1} + Q_{\theta_2}}{2}$, we have:

$$\min(Q_{\theta_1}, Q_{\theta_2}) = \frac{Q_{\theta_1} + Q_{\theta_2}}{2} - \frac{|Q_{\theta_1} - Q_{\theta_2}|}{2} = \bar{Q} - \frac{1}{\sqrt{2}} \underbrace{\sqrt{\frac{(Q_{\theta_1} - \bar{Q})^2 + (Q_{\theta_2} - \bar{Q})^2}{2-1}}}_{\sigma_{\text{epi}}}. \quad (28)$$

Here, the second term is directly identified as the unbiased sample standard deviation σ_{epi} , scaled by a fixed coefficient of $1/\sqrt{2}$. This derivation proves that the minimum operator implicitly constructs a lower confidence bound that accounts solely for epistemic uncertainty while remaining theoretically blind to aleatoric uncertainty.

B.2. Uncertainty Entanglement in DSAC-T

DSAC-T stabilizes training by enforcing a shared target distribution across all ensemble members. Specifically, it identifies the critic with the minimum mean value, denoted by index $\bar{i} = \arg \min_i Q_{\theta_i}(s, a)$, and uses its distribution $Z_{\theta_{\bar{i}}}$ as the regression target for all other critics. We demonstrate here that this mechanism causes the variance head of the non-minimum critic to learn an entangled mixture of aleatoric and epistemic uncertainties.

Consider an ensemble with two critics, θ_1 and θ_2 . Without loss of generality, assume that θ_1 currently provides the minimum mean value (i.e., $\bar{i} = 1$). Consequently, samples $y_{z_{\theta_1}} \sim Z_{\theta_1}$ are used as targets for both networks. For the selected critic θ_1 , the variance head $\sigma_{\theta_1}^2$ is trained to minimize the negative log-likelihood of its own samples. This effectively estimates the variance of the target distribution:

$$\sigma_{\theta_1}^2 \approx \mathbb{E}_{y_{z_{\theta_1}}} [(y_{z_{\theta_1}} - Q_{\theta_1})^2] = \text{Var } Z_{\theta_1}. \quad (29)$$

Here, $\sigma_{\theta_1}^2$ purely captures the aleatoric uncertainty of the current value distribution. However, for the other critic θ_2 , the variance head $\sigma_{\theta_2}^2$ minimizes the residual between the target samples y and its own mean Q_{θ_2} :

$$\begin{aligned} \sigma_{\theta_2}^2 &\approx \mathbb{E}_{y_{z_{\theta_1}}} [(y_{z_{\theta_1}} - Q_{\theta_2})^2] = \mathbb{E}_{y_{z_{\theta_1}}} [(y_{z_{\theta_1}} - Q_{\theta_1} + Q_{\theta_1} - Q_{\theta_2})^2] \\ &= \underbrace{\mathbb{E}_{y_{z_{\theta_1}}} [(y_{z_{\theta_1}} - Q_{\theta_1})^2]}_{\text{Aleatoric uncertainty of } Z_{\theta_1}} + \underbrace{(Q_{\theta_1} - Q_{\theta_2})^2}_{\text{Epistemic uncertainty}} + \underbrace{2(Q_{\theta_1} - Q_{\theta_2})\mathbb{E}_{y_{z_{\theta_1}}} [y_{z_{\theta_1}} - Q_{\theta_1}]}_{=0}. \end{aligned} \quad (30)$$

This derivation reveals two critical flaws in the learning objective. First, the variance $\sigma_{\theta_2}^2$ is explicitly contaminated by the epistemic term $(Q_{\theta_1} - Q_{\theta_2})^2$, meaning the model cannot distinguish between environmental stochasticity and model disagreement. Second, since the index $\bar{i}$ is determined dynamically by the minimum Q-value, it frequently switches between critics during training. Consequently, the regression target for the variance head shifts abruptly and frequently, destabilizing the optimization process while forcing the network to chase a moving target that contains coupled epistemic noise.

B.3. Fallacy in Mean-Embedding Bellman Backups

Accurate uncertainty estimation relies on the correct formulation of the Bellman update target. Here, we analyze the limitations of alternative propagation strategies compared to the full distributional backup employed by DSAC-T and DSAC-AID. Throughout this analysis, we adopt the notation from DSAC, modeling the return distribution as a Gaussian $Z(s, a) \sim \mathcal{N}(Q(s, a), \sigma(s, a))$.

Correct Distributional Backup. As implemented in DSAC-T and DSAC-AID, the correct distributional backup requires the Bellman operator to preserve the stochasticity of future states to accurately learn the return distribution $Z(s, a)$. The target is defined as:

$$\mathcal{T}Z(s, a) = r + \gamma Z(s', a'), \quad \text{where } Z(s', a') \sim \mathcal{N}(Q', \sigma'). \quad (31)$$

By sampling Z' from the next-state distribution, it correctly accumulates uncertainty along the trajectory, capturing both the immediate reward noise and the inherent stochasticity of the future decisions and dynamics.

Fallacious Mean-Embedding Backup. Some algorithms like IV-RL (Mai et al., 2022) and DoubleGum (Hui et al., 2023) simplify the target by replacing the future distribution Z' with its expectation Q' :

$$\mathcal{T}Z(s, a) = r + \gamma Q(s', a') \quad \text{or} \quad \mathcal{T}Q(s, a) = r + \gamma Q(s', a'). \quad (32)$$

By substituting the stochastic sample with a deterministic expectation, this formulation is fundamentally flawed for uncertainty estimation in two respects. First, it fails to capture the full return uncertainty, particularly the long-term

stochasticity beyond the immediate transition. Its variance mainly reflects the stochasticity of the one-step transition, which includes the randomness of r induced by the transition possibility $p(s'|s, a)$ and the variance of Q' induced by $a' \sim \pi$. The uncertainty of future values is partially implicitly contained in Q' , but it is smaller than real uncertainty. This results in a severe underestimation of the return uncertainty. Second, it fails to capture the value uncertainty. Since the mean-embedded target is a single stochastic realization dominated by transition noise, treating its variance as the uncertainty of an expected value is theoretically incorrect. Consequently, this backup target neither correctly models the return distribution nor the value uncertainty. To rigorously estimate the latter, one must employ the correct formulation discussed below.

Correct formulation for direct Q-uncertainty estimation. To correctly learn the uncertainty of the expected value, the target must be based on the expectation of the distribution. The theoretically correct target for Q is:

$$\mathcal{T}Q(s, a) = \mathbb{E}_{s', a'}[r + \gamma Q(s', a')]. \quad (33)$$

In practice, this necessitates sampling multiple next states s' and actions a' to approximate the expectation. This procedure requires multiple forward passes of the policy and value networks for every transition, resulting in prohibitive computational costs. This is why DSAC-AID avoids direct Q-uncertainty learning and instead derives it via the β -weighted mechanism.

C. Algorithmic Implementation Details

C.1. Hyperparameter Decoupling

Decoupled formulation of evaluation penalty. As derived in Eq. (14), the theoretical penalty is $\lambda_{\text{LB}}\sigma_Q$. To prevent λ_{LB} from scaling the aleatoric suppression, we implement:

$$\sigma_Q^{\text{critic}}(s, a) = \sqrt{\sigma_{\text{epi}}^2(s, a) + (\beta/\lambda_{\text{LB}})^2 \sigma_{\text{ale}}^2(s, a)}. \quad (34)$$

In aleatoric-dominated regions ($\sigma_{\text{epi}} \approx 0$), the effective penalty becomes $\lambda_{\text{LB}} \cdot (\beta/\lambda_{\text{LB}})\sigma_{\text{ale}} = \beta\sigma_{\text{ale}}$.

Decoupled formulation of exploration bonus. Similarly, for the exploration bonus $\lambda_{\text{UB}}\sigma_Q$ in Eq. (20), we implement:

$$\sigma_Q^{\text{actor}}(s, a) = \sqrt{\sigma_{\text{epi}}^2(s, a) + (\beta/\lambda_{\text{UB}})^2 \sigma_{\text{ale}}^2(s, a)}. \quad (35)$$

The effective bonus becomes $\lambda_{\text{UB}} \cdot (\beta/\lambda_{\text{UB}})\sigma_{\text{ale}} = \beta\sigma_{\text{ale}}$.

Symmetric cancellation of aleatoric component. By defining the scaling terms inversely to their respective confidence bound coefficient λ , we ensure that the aleatoric component is always weighted by exactly β , regardless of the choices for λ_{LB} and λ_{UB} . Consequently, the optimistic bonus in π_B perfectly cancels the pessimistic penalty in the critic regarding aleatoric uncertainty, while preserving the intended epistemic uncertainty asymmetry.

C.2. Overestimation Metric

The overestimation error at step t is computed as the difference between the current epistemic-penalized lower bound and the discounted return accumulated along a finite trajectory $\{(s_t, a_t, r_t)\}_{t=1}^T$ generated by the pessimistic target policy:

$$e_t = [Q(s_t, a_t) - \lambda_{\text{LB}}\sigma_{\text{epi}}(s_t, a_t)] - \left[\sum_{j=0}^{T-t-1} \gamma^j r_{t+j} - \alpha \sum_{j=1}^{T-t} \gamma^j \log \pi_{t+j} + \gamma^{T-t} [Q(s_T, a_T) - \lambda_{\text{LB}}\sigma_{\text{epi}}(s_T, a_T)] \right]. \quad (36)$$

Limitations. It is crucial to acknowledge an inherent discrepancy in this metric arising from the evaluation protocol. While the Q-value estimate $Q(s, a)$ models the expected return of the *stochastic* policy, the ground truth return is computed via rollout trajectories generated by the *deterministic* evaluation policy. During the early training phase, the deterministic projection of an immature stochastic policy often yields lower returns than the expected return of the stochastic policy, leading to a systematic “false positive” in the overestimation metric e_t . Consequently, the overestimation peak observed in Figure 1 (even for DSAC-AID) partially reflects this measurement artifact rather than a failure of value correction. However, this does not undermine the validity of our analysis regarding aleatoric impulse. Since both DSAC-AID and its variant without aleatoric pessimism are subject to the exact same measurement bias, the observed divergence in their stability profiles is attributable solely to the aleatoric pessimism coefficient β_{init} , rather than the evaluation metric itself.

Advantages. We adhere to this evaluation standard for two practical reasons: (1) *Computational efficiency*: Since RL training routines inherently perform periodic performance evaluations using deterministic policies, calculating bias based on these existing rollouts avoids introducing significant additional computational overhead; and (2) *Variance reduction*: Accurately estimating the true value of a high-entropy stochastic policy suffers from extreme variance. Obtaining a stable ground truth would require averaging a prohibitive number of Monte Carlo trajectories, making it practically infeasible compared to the stable baseline provided by the deterministic mode.

Algorithm 1 DSAC-AID training step with changes with respect to standard SAC colored red.

Require: $\{\theta_i\}_{i=1}^m$ - critic parameters, ϕ_T - target policy parameters, ϕ_B - behavior policy parameters, λ_{LB} - lower confidence bound coefficient, λ_{UB} - upper confidence bound coefficient, α_{init} - temperature, β_{init} - aleatoric pessimism coefficient, l_z - critic learning rate, l_π - actor learning rate, l_α - temperature learning rate, l_β - aleatoric pessimism annealing rate, τ - Polyak coefficient, K - bias estimation sample size

- 1: Initialize target networks: $\bar{\theta}_i \leftarrow \theta_i, \bar{\phi}_T \leftarrow \phi_T$
- 2: **for** each iteration **do**
- 3: **for** each sampling step **do**
- 4: Calculate action using the optimistic behavior policy
- 5: $a \sim \pi_B(a|s)$
- 6: Get reward r and new state s'
- 7: Store samples (s, a, r, s') in buffer $\mathcal{B}$
- 8: **end for**
- 9: **for** each update step **do**
- 10: Sample data from $\mathcal{B}$
- 11: Calculate next action using the pessimistic target policy
- 12: $a' \sim \pi_T(a'|s')$
- 13: Calculate the epistemic and aleatoric uncertainty of total return estimation
- 14: $\sigma_{\text{epi}}^2(s, a) = \frac{1}{m-1} \sum_{i=1}^m [Q_{\bar{\theta}_i}(s, a) - Q(s, a)]^2, \sigma_{\text{ale}}^2(s, a) = \frac{1}{m} \sum_{i=1}^m \sigma_{\bar{\theta}_i}^2(s, a)$
- 15: Calculate the pessimistic targets using the total value uncertainty
- 16: $y_q = r + \gamma [Q(s', a') - \lambda_{LB} \sqrt{\sigma_{\text{epi}}^2(s', a') + (\frac{\beta}{\lambda_{LB}})^2 \sigma_{\text{ale}}^2(s', a')} - \alpha \log \pi_{\bar{\phi}}(a'|s')]$
- 17: Calculate the sampled return targets
- 18: $y_{z_i} = r + \gamma [z'_{\bar{\theta}_i} - \alpha \log \pi_{\bar{\phi}}(a'|s')] | z'_{\bar{\theta}_i} \sim \mathcal{Z}_{\bar{\theta}_i}(\cdot|s', a')$
- 19: Update pessimistic critics
- 20: $\theta_i \leftarrow \theta_i - l_z \mathbb{E}_{\mathcal{B}} \left[- \frac{\mathbb{E}_{\mathcal{B}}[\sigma_{\bar{\theta}_i}^2 + \sigma_{\text{epi}}^2]}{\sigma_{\bar{\theta}_i}^2 + \sigma_{\text{epi}}^2} (y_q - Q_{\theta_i}) \nabla Q_{\theta_i} - \frac{\mathbb{E}_{\mathcal{B}}[\sigma_{\bar{\theta}_i}^2]}{\sigma_{\bar{\theta}_i}^2} \frac{(y_{z_i} - Q_{\theta_i})^2 - \sigma_{\bar{\theta}_i}^2}{\sigma_{\theta_i}} \nabla \sigma_{\theta_i} \right]$
- 21: Update the pessimistic target policy
- 22: $\phi_T \leftarrow \phi_T + l_\pi \mathbb{E}_{s \sim \mathcal{B}, a \sim \pi_T} [Q(s, a) - \lambda_{LB} \sigma_{\text{epi}}(s, a) - \alpha \log(\pi_{\phi_T}(a|s))]$
- 23: Update the optimistic behavior policy
- 24: $\phi_B \leftarrow \phi_B + l_\pi \mathbb{E}_{s \sim \mathcal{B}, a \sim \pi_B} [Q(s, a) + \lambda_{UB} \sqrt{\sigma_{\text{epi}}^2(s, a) + (\frac{\beta}{\lambda_{UB}})^2 \sigma_{\text{ale}}^2(s, a)} - \alpha \log(\pi_{\phi_B}(a|s))]$
- 25: Update entropy temperature
- 26: $\alpha \leftarrow \alpha - l_\alpha \mathbb{E}_{a \sim \pi_{\phi_T}} [-\log \pi_{\phi_T}(a|s) - \bar{\mathcal{H}}]$
- 27: Update target networks
- 28: $\bar{\theta}_i \leftarrow \tau \theta_i + (1 - \tau) \bar{\theta}_i, \bar{\phi}_T \leftarrow \tau \phi_T + (1 - \tau) \bar{\phi}_T$
- 29: **end for**
- 30: **if** evaluation phase **then**
- 31: Collect evaluation trajectories $\{(s_t, a_t, r_t)\}_{t=1}^T$ using the pessimistic target policy
- 32: Calculate total average return (TAR)
- 33: Calculate raw overestimation errors using (36)
- 34: Update aleatoric pessimism coefficient β using relative overestimation errors (ROE)
- 35: $\beta \leftarrow \beta + l_\beta \frac{\min(0, \frac{1}{K} \sum_{t=1}^K e_t(s_t, a_t))}{\mathbb{E}_{(s, a) \sim \mathcal{B}} [\sigma_Q(s, a)]}$
- 36: **end if**
- 37: **end for**

C.3. Complete Pseudocode

We present the complete pseudocode for DSAC-AID in Algorithm 1. Our method is built upon the distributional soft actor-critic framework. To compare with standard SAC, we highlight the core mechanisms introduced by AID in **red**.

D. Experimental Details

D.1. Benchmark Description

DeepMind Control Suite. We evaluate on the 7 most challenging tasks involving the dog and humanoid agents. The dog agent has a 38-dimensional continuous action space and 223-dimensional state space, while the humanoid agent has a 24-dimensional continuous action space and 67-dimensional state space. These tasks fall into two categories: (1) *Standing tasks*, where the agent must maintain upright balance, and (2) *Moving tasks*, which additionally require the agent to move at a target velocity. For moving tasks, the reward is defined as the product of the standing reward and the forward velocity reward, i.e., $\text{Reward} = (\text{Standing reward}) \times (\text{Forward velocity reward})$.



Figure 8. Dog (DMC)

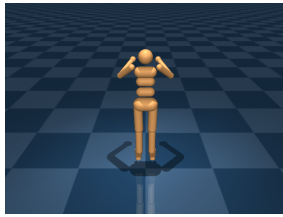


Figure 9. Humanoid (DMC)

Standing reward: Encourages the agent to maintain an upright posture.

Forward velocity reward: Ensures the agent moves at the target speed (1 m/s for dog-walk, 3 m/s for dog-trot, 9 m/s for dog-run, 1 m/s for humanoid-walk and 10 m/s for humanoid-run).

MuJoCo Continuous Control. We further evaluate the algorithms on high-dimensional locomotion tasks from the OpenAI Gym MuJoCo suite, specifically focusing on the most complex quadrupedal and bipedal agents.

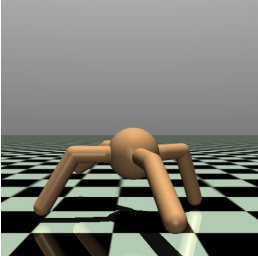


Figure 10. Ant (Gym)

Ant-v3: The agent controls a four-legged quadruped (111-dim state, 8-dim action) to move forward as fast as possible. The challenge lies in coordinating high-dimensional actions for stable gait generation.

Reward: $r = v + r_{\text{healthy}} - C_{\text{ctrl}} - C_{\text{contact}}$.

The reward incentivizes forward velocity (v) and maintaining an upright posture (r_{healthy}), while penalizing large control inputs (C_{ctrl}) and excessive external contact forces (C_{contact}).



Figure 11. Humanoid (Gym)

Humanoid Domain: A bipedal robot (376-dim state, 17-dim action) evaluated on two distinct tasks:

1. Humanoid-v3: The goal is to maintain balance and move forward.

Reward: $r = 1.25v + r_{\text{healthy}} - C_{\text{ctrl}} - C_{\text{contact}}$. It heavily incentivizes forward velocity (v) and maintaining an upright posture (r_{healthy}) while penalizing control costs and excessive external contact forces (C_{contact}).

2. HumanoidStandup-v4: The goal is to stand up from a random lying position.

Reward: $r = h_{\text{head}} + r_{\text{healthy}} - C_{\text{ctrl}} - C_{\text{contact}}$. It focuses on maximizing head height (h_{head}) rather than horizontal movement.

D.2. Implementation Details

Implementation framework. DSAC-AID is implemented along with standard baselines including PPO, SAC, and DSAC-T within the GOPS framework (Wang et al., 2023), an open-source PyTorch-based reinforcement learning solver. For a fair comparison against state-of-the-art methods, the official implementations of DoubleGum and BRO were utilized directly from their respective GitHub repositories. A detailed summary of the codebases is provided in Table 2.

Table 2. Summary of Implementation Frameworks and Code Repositories.

Algorithm / Framework	Source Repository
GOPS (Wang et al., 2023)	https://github.com/Intelligent-Driving-Laboratory/GOPS
DSAC-AID (Ours)	https://github.com/yzy-yuzhouyang/DSAC-AID
DoubleGum (Hui et al., 2023)	https://github.com/dyth/doublegum
BRO (Nauman et al., 2024)	https://github.com/naumix/BiggerRegularizedOptimistic

Network architecture. PPO, SAC, and DSAC-T utilize simple multi-layer perceptrons (MLPs). DSAC-AID adopts DoubleGum’s robust architecture using MLPs with orthogonal initialization (Saxe et al., 2013) and GroupNorm (Wu & He, 2018) without a shift and scale (Xu et al., 2019), except on Humanoid-v3 and Ant-v3. Distinct from these compact networks, BRO (Nauman et al., 2024) retains its original, significantly larger network architecture, which is intrinsic to its design for heavy structural regularization. The model-based baselines additionally learn a world model to capture environment dynamics, with TD-MPC2 and its variants (BMPC and BOOM) specifically employing LayerNorm (Ba et al., 2016).

Table 3. General hyperparameters for model-free algorithms. v3 *MuJoCo* refers specifically to the legacy Ant-v3 and Humanoid-v3 tasks. *Others* encompasses all DMC tasks and the more recent HumanoidStandup-v4 task.

Hyperparameter	DSAC-AID (Ours)	DSAC-T	SAC	DoubleGum	BRO	PPO
Common						
Discount Factor (γ)	0.99	0.99	0.99	0.99	0.99	-
Reward Scale (MuJoCo / DMC)	0.1 / 50	0.1 / 50	0.1 / 50	1 / 1	1 / 1	1 / 1
Replay Buffer & Batch Settings						
Replay Batch Size	256	256	256	256	256	2000
Sample Batch Size	20	20	20	20	1	2000
Replay Buffer Size	10^6	10^6	10^6	10^6	10^6	-
Warmup Steps	10,000	10,000	10,000	10,000	10,000	-
Replay Ratio	1	1	1	1	10	-
Polyak (τ)	0.005	-	0.005	0.005	0.005	-
Neural Network						
Critic Ensemble (Online/Target)	4 / 4	2 / 2	2 / 2	2 / 2	2 / 2	1 / -
Critic Hidden (Depth $\times$ Dim)	3×256	3×256	3×256	3×256	5×512	2×64
Actor Ensemble (Online/Target)	2 / 1	1 / 1	1 / -	1 / 1	2 / -	1 / -
Actor Hidden (Depth $\times$ Dim)	3×256	3×256	3×256	3×256	1×256	2×64
Activation (v3 MuJoCo / Others)	GeLU / ReLU	GeLU / GeLU	GeLU / GeLU	ReLU / ReLU	ReLU / ReLU	Tanh / Tanh
Normalization (v3 MuJoCo / Others)	- / GroupNorm	-	-	GroupNorm	LayerNorm	-
Learning Rate (v3 MuJoCo / Others)	1×10^{-4} / 3×10^{-4}	1×10^{-4}	1×10^{-4}	3×10^{-4}	3×10^{-4}	3×10^{-4}
Total Parameters	1.71M	0.93M	0.78M	0.93M	4.93M	0.02M
Entropy						
Target Entropy ($\bar{\mathcal{H}}$)	$- \mathcal{A} $	$- \mathcal{A} $	$- \mathcal{A} $	-	$- \mathcal{A} /2$	-
Initial Alpha (α)	2.72	2.72	2.72	-	1.00	-
α Learning Rate	3×10^{-4}	3×10^{-4}	3×10^{-4}	-	3×10^{-4}	-

Hyperparameters. A comprehensive comparison of the general hyperparameter settings for all evaluated algorithms is summarized in Table 3. To ensure a fair comparison, we standardized the configurations where appropriate. For DoubleGum, we aligned its network architecture and sampling batch size with standard-capacity baselines (e.g., SAC and DSAC-T). This alignment eliminates advantages arising purely from model capacity. Consequently, its pessimism coefficient was fixed at $c = -0.1$, adhering to the recommended configuration. Regarding BRO, while we extended the total training iterations to match our evaluation benchmark, we retained its original sampling configuration. This is because our preliminary experiments indicated that imposing the standard batch size on BRO resulted in significant performance degradation. Finally, for our proposed DSAC-AID, the specific hyperparameters governing the AID mechanism are detailed in Table 4. Notably, for HumanoidStandup-v4, the aleatoric pessimism is deactivated, as the value estimates in this task exhibit no distinct overestimation. Conversely, for challenging tasks like DMC Humanoid, we reduce the effective epistemic coefficient λ to prevent excessive pessimism when coupled with network regularization, yet retain full aleatoric pessimism. This distinction underscores that network regularization remains ineffective against the specific instability driven by aleatoric uncertainty.

Table 4. Task-specific hyperparameters for DSAC-AID.

Benchmark	Task	Confidence Bound Coefficients		Aleatoric Pessimism Coefficient (β)			
		λ_{LB} (Lower)	λ_{UB} (Upper)	Initial	Annealing Rate	Update Interval	K
MuJoCo	Ant-v3	0.5	0.5	0.03	1.5×10^{-4}	2500	200
	Humanoid-v3	1.0	1.0	0.20	5.0×10^{-4}	2500	200
	HumanoidStandup-v4	0.5	0.5	0.00	-	-	200
DMC	Dog tasks	$1/\sqrt{2}$	$1/\sqrt{2}$	0.10	5.0×10^{-3}	10000	200
	Humanoid-stand	0.1	0.1	0.10	5.0×10^{-3}	10000	200
	Humanoid-walk	0.1	0.1	0.10	2.0×10^{-3}	10000	200
	Humanoid-run	1×10^{-4}	1×10^{-4}	0.10	2.0×10^{-3}	10000	200

E. Extended Experimental Results

E.1. Comparison with Model-based Baselines

Model-based reinforcement learning (MBRL) is widely recognized for its ability to significantly enhance sample efficiency by learning explicit world models. Representative approaches, such as DreamerV3, achieve this by leveraging latent imagination to simulate future trajectories. Crucially, incorporating off-policy capabilities further amplifies this efficiency by facilitating the reuse of historical datasets generated by diverse behavior policies (Guan et al., 2021). This paradigm is exemplified by methods like TD-MPC2 and its subsequent improvements (BMPC and BOOM), which maintain an off-policy core for their critic updates while utilizing world models to perform online planning.

To provide a comprehensive evaluation for DSAC-AID, we directly utilize the training curves and numerical results reported in the BOOM paper (Zhan et al., 2025). Notably, we align the comparison based on iteration steps rather than environment steps. While model-based methods like BOOM ostensibly require fewer environment interactions, they necessitate extensive interactions with the learned world model, which are computationally expensive but often excluded from sample-efficiency counts. In practice, DSAC-AID consumes comparable wall-clock time to BOOM for the same number of iterations. Thus, this iteration-based alignment offers a fair assessment of total training effort. As shown in Figure 12 and Table 5, remarkably, even without the support of online planning or explicit world models, DSAC-AID surpasses the previous state-of-the-art BOOM by an average margin of **5.5%**. This superiority is particularly pronounced in the most challenging tasks involving high-dimensional dynamics: compared to BOOM, DSAC-AID achieves substantial performance gains of **7.7%** on Dog-run and **17.7%** on Humanoid-run.

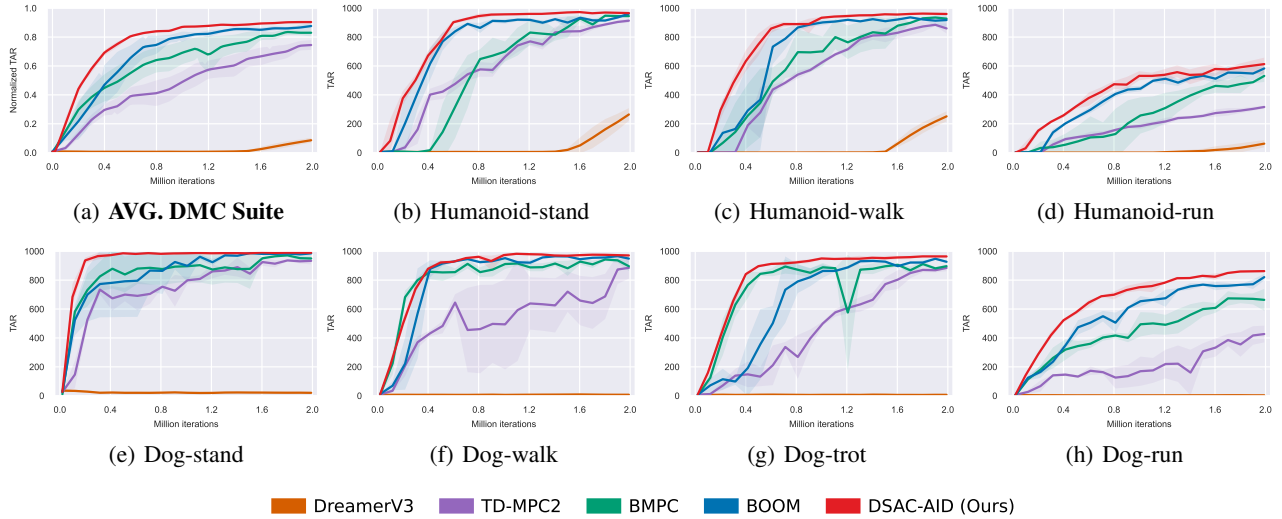


Figure 12. Training curves on benchmarks comparing DSAC-AID with model-based algorithms. The solid lines represent the mean, while the shaded regions indicate the confidence interval over 3 seeds. The average performance curve appears at the left of the 1st row, which is normalized by 1k before averaging.

Table 5. Total Average Return (TAR) on DMC Suite tasks comparing DSAC-AID with model-based algorithms. Mean $\pm$ Std over 3 seeds. Darker Blue = best, Lighter Blue = second-best. The dagger symbol ($\dagger$) denotes scores normalized by 1k before averaging.

Task	DreamerV3	TD-MPC2	BMPC	BOOM	DSAC-AID (ours)
Dog-stand	35.4 $\pm$ 10.8	936.4 $\pm$ 7.6	971.3 $\pm$ 11.0	986.8 $\pm$ 1.8	994.6 $\pm$ 0.3
Dog-walk	9.1 $\pm$ 0.6	885.0 $\pm$ 74.8	942.9 $\pm$ 9.6	965.4 $\pm$ 0.3	991.3 $\pm$ 1.9
Dog-trot	7.9 $\pm$ 0.7	884.4 $\pm$ 22.2	911.3 $\pm$ 18.2	947.9 $\pm$ 4.7	982.4 $\pm$ 3.8
Dog-run	4.3 $\pm$ 3.2	427.0 $\pm$ 57.9	673.7 $\pm$ 50.2	820.7 $\pm$ 23.0	884.2 $\pm$ 4.4
Humanoid-stand	264.5 $\pm$ 44.1	913.3 $\pm$ 14.7	947.9 $\pm$ 4.4	962.1 $\pm$ 10.7	976.1 $\pm$ 1.1
Humanoid-walk	251.5 $\pm$ 35.2	884.8 $\pm$ 8.3	935.1 $\pm$ 4.0	936.1 $\pm$ 3.3	967.2 $\pm$ 0.8
Humanoid-run	62.5 $\pm$ 25.8	316.2 $\pm$ 9.2	531.2 $\pm$ 42.0	582.8 $\pm$ 26.0	685.9 $\pm$ 28.3
AVG. DMC Suite †	0.088 $\pm$ 0.110	0.746 $\pm$ 0.244	0.836 $\pm$ 0.161	0.878 $\pm$ 0.134	0.926 $\pm$ 0.105

E.2. Performance with Minimal Ensemble Size

While the default DSAC-AID utilizes $m = 4$ critics to provide a statistically richer approximation of the posterior distribution, many foundational off-policy baselines such as SAC, TD3, and BRO rely on a twin-critic ($m = 2$) architecture. To further evaluate the efficiency and robustness of the AID mechanism, we investigate the performance of DSAC-AID under a minimal ensemble setting ($m = 2$).

As demonstrated in Table 7, reducing the ensemble size to $m = 2$ significantly decreases the computational overhead, bringing the training time to 1.75x relative to SAC, compared to 2.35x for the default configuration. Remarkably, even with only two critics, DSAC-AID consistently outperforms state-of-the-art model-free methods (BRO, DoubleGum) and the leading model-based approach (BOOM) across all high-dimensional tasks. This suggests that the core of our performance gain stems from the precise uncertainty disentanglement and damping of aleatoric impulse, rather than a simple increase in model capacity.

Table 6. Comparison of DSAC-AID with minimal ensemble size ($m = 2$) against state-of-the-art baselines. Gym-MuJoCo values are in k (10^3). Mean $\pm$ Std over 3 seeds. Darker Blue = best, Lighter Blue = second-best. The average scores of individual benchmarks ($\dagger$) are normalized by task-specific thresholds before averaging: Ant (10k), Humanoid (15k), and HumanoidStandup (450k) for Gym-MuJoCo; all DMC Suite tasks are normalized by 1k.

Task	BRO	DoubleGum	BOOM	DSAC-AID ($m = 2$ & $m = 4$)	
Ant-v3 (k)	8.3 $\pm$ 0.1	7.5 $\pm$ 0.0	-	9.8 $\pm$ 0.5	10.0 $\pm$ 0.4
Humanoid-v3 (k)	11.0 $\pm$ 0.3	10.8 $\pm$ 0.9	-	11.8 $\pm$ 0.2	12.7 $\pm$ 0.3
HumanoidStandup-v4 (k)	368.4 $\pm$ 46.0	252.5 $\pm$ 37.0	-	401.3 $\pm$ 2.1	405.2 $\pm$ 1.9
AVG. Gym-MuJoCo †	0.793 $\pm$ 0.074	0.677 $\pm$ 0.101	-	0.885 $\pm$ 0.081	0.918 $\pm$ 0.070
Dog-stand	993.6 $\pm$ 4.0	992.7 $\pm$ 1.8	986.8 $\pm$ 1.1	995.7 $\pm$ 0.8	994.6 $\pm$ 0.3
Dog-walk	964.2 $\pm$ 1.7	965.4 $\pm$ 7.9	965.4 $\pm$ 0.3	992.4 $\pm$ 2.1	991.3 $\pm$ 1.9
Dog-trot	944.7 $\pm$ 2.2	907.2 $\pm$ 25.1	947.9 $\pm$ 4.7	962.2 $\pm$ 5.8	982.4 $\pm$ 3.8
Dog-run	687.4 $\pm$ 15.9	699.6 $\pm$ 37.1	820.7 $\pm$ 33.0	850.6 $\pm$ 4.1	884.2 $\pm$ 4.4
Humanoid-stand	961.8 $\pm$ 4.7	966.7 $\pm$ 3.2	962.1 $\pm$ 10.7	974.0 $\pm$ 1.6	976.1 $\pm$ 1.1
Humanoid-walk	940.3 $\pm$ 6.7	951.1 $\pm$ 5.7	936.1 $\pm$ 7.3	967.1 $\pm$ 1.1	967.2 $\pm$ 0.8
Humanoid-run	472.4 $\pm$ 66.3	523.6 $\pm$ 42.1	582.8 $\pm$ 20.0	606.9 $\pm$ 63.8	685.3 $\pm$ 28.2
AVG. DMC Suite †	0.853 $\pm$ 0.181	0.858 $\pm$ 0.166	0.878 $\pm$ 0.134	0.907 $\pm$ 0.133	0.926 $\pm$ 0.105

E.3. Extended Analysis of Aleatoric Impulse

To investigate the prevalence and impact of aleatoric impulse, we analyzed the value estimation bias of various state-of-the-art model-free off-policy algorithms on the complex Humanoid-v3 task. To ensure a fair comparison, we standardized the reward scale to 0.1 for BRO and DoubleGum (yielding no improvement in TAR). The comparative results, depicting both the full training trajectory and the early-stage dynamics, are presented in Figure 13.

Objective-regularized baselines (SAC, DSAC-T). Both SAC and DSAC-T exhibit significant overestimation peaks during the initial phase. This behavior is expected, as neither algorithm employs layer normalization nor explicitly suppresses

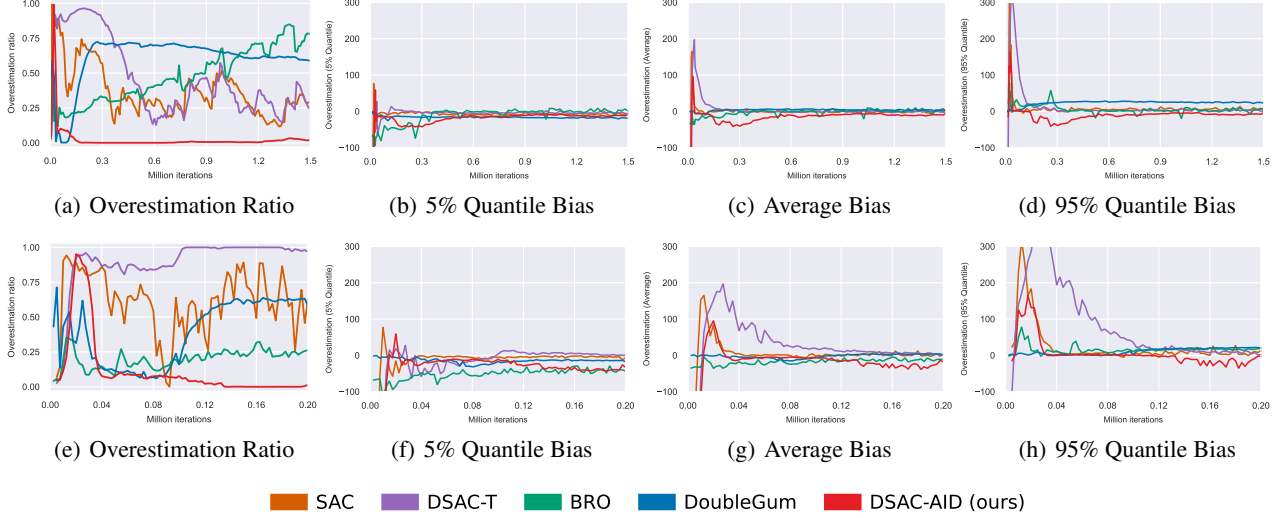


Figure 13. Analysis of value overestimation on the Humanoid-v3 comparing DSAC-AID with baseline algorithms. The top row shows the evolution of these metrics over the full training course (0–1.5M iterations), illustrating long-term convergence properties. The bottom row provides a zoomed-in view of the initial phase (0–0.2M iterations) to highlight the transient aleatoric impulse phenomenon.

aleatoric impulse. Consequently, the high initial entropy combined with sparse data leads to severe aleatoric impulse. Part of this overestimation is attributable to the measurement bias discussed in Appendix C.2.

Network-regularized baselines (BRO, DoubleGum). By applying layer normalization, both methods effectively mitigate the average bias, with DoubleGum further benefiting from its deterministic nature to avoid entropy-induced metric inaccuracies. However, neither method effectively constrains estimation variance. As evidenced by the wide gap between the 5% and 95% quantiles, this high variance manifests as distinct peaks in the overestimation ratio. These results demonstrate that while network regularization can stabilize the average error, it lacks the granularity to specifically suppress the transient surges driven by aleatoric impulse. We conjecture that network regularization primarily filters high-frequency noise rooted in estimation errors (epistemic uncertainty), while leaving the low-frequency variance arising from inherent stochasticity (aleatoric uncertainty) unconstrained.

DSAC-AID (ours). In contrast, DSAC-AID achieves the lowest overestimation ratio among all compared methods, notably without relying on layer normalization for this task. This superior stability is directly attributable to the AID mechanism. Furthermore, the adaptive regulation of β ensures that the injected pessimism is transient. As shown in the top row of Figure 13, the estimation bias converges to a neutral state in the later stage, ensuring that the initial damping does not hinder long-term policy learning.

E.4. Visualization of Uncertainty Dynamics

Figure 14 visualizes the uncertainty evolution on Humanoid-v3, where the estimated aleatoric uncertainty faithfully monitors the intrinsic stochasticity, including the early impulse. This alignment justifies its utility as a reliable metric for the penalty term. Leveraging this, the aleatoric coefficient β effectively acts as a temporal gate, imposing heavy aleatoric penalization during this initial volatility before annealing towards zero. Consequently, the constructed total value uncertainty transitions from an aleatoric-dominated regime to a purely epistemic one. This dynamic regulation ensures that the penalty mechanism specifically targets the transient surge without hindering long-term value estimation.

E.5. Evaluation of Computational Efficiency

To evaluate the computational cost of DSAC-AID, we record the wall-clock training time against a suite of representative off-policy model-free algorithms on the high-dimensional Humanoid-run task. All algorithms were trained for 2×10^6 steps under identical hardware and software environments to ensure a fair comparison.

Hardware specifications. All experiments were conducted on a single device. The hardware configuration consists of an AMD EPYC 7402 24-Core Processor and an NVIDIA GeForce RTX 4090 GPU.

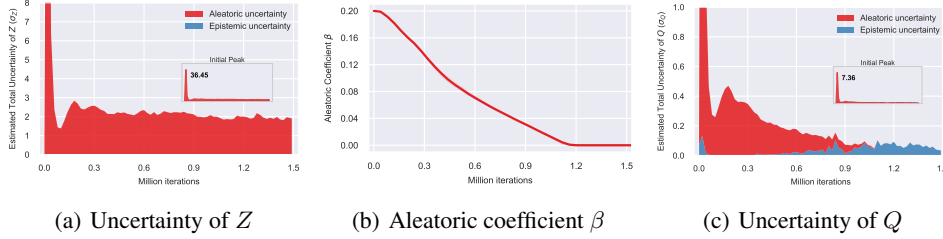


Figure 14. **Evolution of uncertainty components during training.** (a) The total uncertainty of the distributional value Z is dominated by the aleatoric component (red), which exhibits a massive peak in the early phase. (b) The annealing curve of β , which serves as the weighting coefficient for the aleatoric component in the value uncertainty estimation. (c) The uncertainty composition of the value Q .

Training time comparison. Table 7 details the total training duration for each algorithm. We also include an ablation on the ensemble size m for DSAC-AID to highlight the trade-off between estimation accuracy and computational cost.

Table 7. Comparison of wall-clock training time against off-policy model-free baselines on Humanoid-run (2×10^6 steps).

Algorithm	Ensemble Size	Time (Hours)	Relative to SAC
SAC	2	15.55	1.00×
DSAC-T	2	21.70	1.40×
BRO	2	24.13	1.55×
DoubleGum	2	47.78	3.07×
DSAC-AID ($m = 2$)	2	27.24	1.75×
DSAC-AID (Default)	4	36.59	2.35×
DSAC-AID ($m = 6$)	6	46.02	2.96×

As shown in Table 7, DSAC-AID significantly outperforms DoubleGum by leveraging vectorized sampling across 4 parallel environments. However, it remains slower than BRO, primarily because BRO benefits from JAX’s compilation optimization while DSAC-AID is implemented in standard PyTorch. This framework difference represents a clear avenue for future efficiency optimization.

F. Extended Related Work on Objective Regularization

To contextualize our contributions, we provide a systematic comparison of DSAC-AID against representative RL algorithms with objective regularization in Table 8. We analyze these methods across three critical dimensions: uncertainty modeling, utilization mechanisms, and coefficient regulation strategies.

F.1. Uncertainty Modeling

The uncertainty in reinforcement learning can be divided into four categories: epistemic uncertainty, aleatoric uncertainty, the total return uncertainty, and the total value uncertainty.

Epistemic uncertainty. Regarding parameter estimation error, scalable ensembles ($m > 2$) (Lee et al., 2021; Li et al., 2023) provide a statistically richer approximation of the posterior distribution compared to the rudimentary Double Q-learning mechanism (Van Hasselt et al., 2016; Haarnoja et al., 2018), though this increases computational cost.

Aleatoric uncertainty and total return uncertainty. Regarding the capture of intrinsic environmental stochasticity, approaches typically fall into two categories. Discrete quantile approaches (Kuznetsov et al., 2020; Clements et al., 2019; Moskovitz et al., 2021) operate without rigid parametric assumptions, retaining higher degrees of freedom to capture complex, multimodal return distributions in highly stochastic settings. Conversely, modeling the full continuous Gaussian distribution (Hui et al., 2023; Mai et al., 2022; Duan et al., 2025) offers superior analytical tractability and precision, particularly in deterministic or low-variance environments.

Total value uncertainty. Our DSAC-AID is the first algorithm to explicitly model the total value uncertainty by disen-

Table 8. Comparison of reinforcement learning algorithms with objective regularization. **Modeling:** *Ens* (Ensemble), *Dbl* (Double Q), *Gauss* (Gaussian), *Quant* (Quantile). **Utilization:** *GM* = Gradient Modulation, *PT* = Pessimistic Target, *OE* = Optimistic Exploration. Background colors denote *explicit* injection mechanisms: *Epi.* (σ_{epi}^2), *Ale.* (σ_{ale}^2), *Total Z* ($\sigma_{\text{epi}}^2 + \sigma_{\text{ale}}^2$), and *Total Q* ($\sigma_{\text{epi}}^2 + \beta\sigma_{\text{ale}}^2$). Unshaded cells denote *implicit* mechanisms. **Coefficient:** *Fixed* (Fixed), *Meta* (Meta-optimization), *Closed* (Closed-loop Regulation).

Algorithm	Uncertainty Modeling		Utilization Mechanism			Coefficient	
	Epi.	Ale./Total	GM	PT	OE	Epi.	Ale./Total
SAC (Haarnoja et al., 2018)	Dbl	–	–	Epi.	–	Fixed	–
TD3 (Fujimoto et al., 2018)	Dbl	–	–	Epi.	–	Fixed	–
OAC (Ciosek et al., 2019)	Dbl	–	–	Epi.	Epi.	Fixed	–
WD3 (He & Hou, 2020)	Dbl	–	–	Epi.	–	Fixed	–
REDQ (Chen et al., 2021)	Ens	–	–	Epi.	–	Fixed	–
Maxmin (Lan et al., 2020)	Ens	–	–	Epi.	–	Fixed	–
SUNRISE (Lee et al., 2021)	Ens	–	Epi.	–	Epi.	Fixed	–
RAC (Li et al., 2023)	Ens	–	–	Epi.	–	Fixed	–
GPL (Cetin & Celiktutan, 2023)	Ens	–	–	Epi.	–	Closed	–
TQC (Kuznetsov et al., 2020)	Ens	Quant	–	Total Z	–	Fixed	Fixed
UA-DQN (Clements et al., 2019)	Dbl	Quant	–	Ale.	Epi.	Fixed	Fixed
TOP (Moskovitz et al., 2021)	Dbl	Quant	–	Epi.	Epi.	Meta	–
BRO (Nauman et al., 2024)	Dbl	Quant	–	–	Epi.	Closed	–
IV-RL (Mai et al., 2022)	Ens	Gauss	Total Z	–	–	–	–
DoubleGum (Hui et al., 2023)	Dbl	Gauss	Total Z	Total Z	–	Fixed	–
DSAC-T (Duan et al., 2025)	Dbl	Gauss	Total Z	Epi.	–	Fixed	–
DSAC-AID (Ours)	Ens	Gauss	Total Z	Total Q	Total Q	Fixed	Closed

gling the return distribution into epistemic and aleatoric components, followed by their adaptive weighted recombination.

F.2. Uncertainty Utilization

Uncertainty information in RL generally serves three distinct purposes: (1) Gradient modulation to stabilize optimization, (2) Pessimistic evaluation to mitigate overestimation, and (3) Optimistic exploration.

Gradient modulation. SUNRISE (Lee et al., 2021) utilizes uncertainty-weighted regression to down-weight uncertain samples, yet restricts this modulation solely to epistemic uncertainty. Conversely, methods such as DoubleGum (Hui et al., 2023) and IV-RL (Mai et al., 2022) explicitly scale gradients based on the estimated total return uncertainty. However, these approaches deviate from the strict distributional Bellman operator, resulting in learned variances that do not faithfully represent the true return uncertainty (refer to Appendix B.3). In contrast, DSAC-T (Duan et al., 2025) minimizes the KL divergence between the distributional Bellman target and the parametric return distribution, providing rigorous theoretical guarantees for uncertainty estimation. Aligning with this strict paradigm, DSAC-AID adopts the reconstructed total variance for gradient modulation, thereby inheriting the robust representation learning validated in DSAC-T.

Pessimistic target. Foundational algorithms like SAC (Haarnoja et al., 2018) and TD3 (Fujimoto et al., 2018) rely on the minimization of double critics to impose *implicit* penalties. Crucially, this heuristic imposes an uncontrolled epistemic penalty, lacking the granularity to explicitly regulate the degree of pessimism. To address this rigidity, Maxmin (Lan et al., 2020) enables control via ensemble size but strictly couples penalty strength with computational cost. REDQ (Chen et al., 2021) decouples these by taking the minimum over a random subset, yet it fails to fully utilize the information from the entire ensemble. TQC (Kuznetsov et al., 2020) implicitly penalizes both uncertainties by truncating the return distribution, but strictly speaking, it penalizes the total return uncertainty rather than the total value uncertainty.

In the realm of *explicit* pessimism, RAC (Li et al., 2023) and GPL (Cetin & Celiktutan, 2023) restrict penalties to epistemic uncertainty, whereas UA-DQN (Clements et al., 2019) focuses solely on aleatoric uncertainty. While DoubleGum (Hui et al., 2023) incorporates both sources, it erroneously penalizes the total return uncertainty. In contrast, DSAC-AID explicitly disentangles these sources to construct a dynamic, closed-loop penalty targeting the precise total value uncertainty.

Optimistic exploration. Exploration strategies exemplified by OAC (Ciosek et al., 2019) typically incorporate an epistemic bonus into the value estimate. This design is consistent with their objective formulation, which does not penalize aleatoric

uncertainty in the critic. In contrast, since DSAC-AID explicitly suppresses aleatoric uncertainty in the critic to mitigate risk, it necessitates a symmetric compensation mechanism to prevent excessive conservatism. Consequently, our framework utilizes the total value uncertainty as the exploration bonus, maintaining active exploration in stochastic regions.

In summary, DSAC-AID distinguishes itself as the only framework that synergizes all three mechanisms. By explicitly decoupling and reconstructing uncertainty components, it rectifies the theoretical misalignment found in prior works, effectively realigning the focus of the pessimistic penalty from the total return uncertainty to the total value uncertainty.

F.3. Adaptive Coefficient Regulation

The effectiveness of uncertainty bounds relies on precise coefficient regulation. Existing dynamic methods, such as TOP (Moskovitz et al., 2021), typically rely on performance-driven meta-optimization. This approach is risky: if the policy is converging towards a local optimum due to overestimation, the performance feedback becomes misleading, potentially reinforcing failure modes. In contrast, DSAC-AID employs a closed-loop regulation mechanism driven directly by the overestimation bias. This establishes a robust negative feedback loop that rectifies bias independent of current policy quality, ensuring stability where performance-dependent meta-gradients fail.